

MMPC-019
TOTAL QUALITY
MANAGEMENT**Block****2****STRATEGIC PERSPECTIVES**

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BLOCK 2 STRATEGIC PERSPECTIVES

This block has following three units which presents a discussion on strategic considerations in TQM. It examines the economics of quality. The emphasis is on the linkage between TQM and business strategy and the elements of quality-centered strategic planning are discussed.

Unit 4 Economics of Quality

Unit 5 TQM and Business Strategy

Unit 6 Quality Centred Strategic Planning



UNIT 4 ECONOMICS OF QUALITY

Objectives

After reading this unit you should be able to develop an understanding and appreciation of:

- the benefits and costs of quality;
- various types of quality-related costs;
- the need for a quality cost system;
- the link between quality and productivity;
- some of the useful techniques of budgeting, control and reporting of quality costs;
- the perspectives on quality costs; and
- the uses of quality cost information.

Structure

- 4.1 Introduction
- 4.2 Quality related Benefits
- 4.3 Quality-related Costs
- 4.4 Life Cycle Costs
- 4.5 Quality and Productivity
- 4.6 Approaches to Quality Costs
- 4.7 Quality Costing and Cost System
- 4.8 Uses of Quality Cost Information
- 4.9 Summary
- 4.10 Key Words
- 4.11 Self-Assessment Questions
- 4.12 Further Readings

4.1 INTRODUCTION

In this unit we will discuss an important strategic aspect that relates to ‘economics of quality’. Whenever we talk about economics of quality, two things come to our mind i.e. what are the benefits, and what are the costs of quality? In other words, what are the benefits and costs involved in introducing, adopting or implementing a quality system? We shall discuss the benefits of a quality system briefly. ‘Cost of quality’ is the main focus of this unit. Nevertheless, we shall begin with a recapitulation of benefits of quality. We will discuss the various types of quality costs. We will also explore the relationship between quality and productivity and also highlight the need for

quality cost system. Thereafter we will deal with the budgeting, control and reporting of quality costs, and discuss some of the techniques that could be useful in this context. Towards the end we shall explain some of the uses of quality cost information.

4.2 QUALITY-RELATED BENEFITS

Quality leads to several benefits for the organization is no longer a matter of dispute. If the customers perceive that the product of a particular organization is relatively of better quality than the products of the competitors, this in itself is a single most important point that may affect the long-term performance of the organization. While there are several benefits which can be quantified, there are several others which are hard to quantify. However, it is generally concluded that 'quality-related benefits' reach far beyond those which are calculable from immediate data relating to success or failure in conformance to specific quality criteria.

Many studies have been done examining the relationship between quality, productivity and marketshare. The unequivocal conclusion of the studies is that quality drives market share. When superior quality and market share are both present, profitability is virtually guaranteed. There is a strong relationship between perceived quality and profitability. Quality pushes growth. It can reduce costs which provide a competitive edge. There are two types of quality: customer-driven (or external) quality, and conformance (or internal specification) quality. The latter relates to appropriate product specifications and service standards that lead to cost reduction. There is an inverse relationship between internal or conformance quality and costs. That is why perhaps Crosby coined the phrase "Quality is free", as quality improves, the cost of the conformance improves, resulting in improved market share and hence profitability and growth.

This, in turn, provides a stimulus for further investment in quality improvement areas such as research and development. The cycle thus goes on. In short, improving both internal (conformance) quality and external (customer perceived) quality not only lowers the cost of poor quality or no quality, but also serves as a driver for growth, market share and profitability. The rewards of higher quality have been found to be positive, substantive and pervasive. The following organizational benefits are often attributed to a quality improvement programme:

- Increased market share
- Greater customer loyalty i.e. customer retention
- Higher (stock market) or share prices
- Higher prices for product services
- Greater productivity
- Accelerated development of new products and their introduction in the market.
- Enhanced employee morale and retention

- Cost-effective use of resources (cost reduction)
- Improved profitability

If TQM programmes do not make significant impact on or are not followed by tangible results in terms of the competitiveness of the organization, the fault must lie somewhere else, and not with the programme per se. An important fact to recognise is that in general the benefits do not and cannot happen overnight. Evidence suggests that achieving success is a slow process. Japanese organizations such as Honda, Nissan and Toyota, which are acknowledged leaders in quality management, have had a long experience (of more than 30 years or so) in implementing quality approach.

Employee empowerment is one of the characteristics of a successful TQM programme. It is a well-known fact that quality-related benefits arise from consumer-based transactions. However, benefits may also flow from transactions between organizations, and from transactions within an organization. According to the transactional model, where buyer and seller are involved in a transaction, quality is a subject of bargaining process in which the other parameters are delivery and cost. This process forms a part of what has been described as an 'adversarial' relationship between buyer and seller, a process in which competitive tendering on pricing and dealing at arm's length with more than one (rather many) suppliers is often a feature. But changing times have necessitated a reappraisal of this traditional model. Gradually, a realisation has dawned on the organizations that they cannot be competitive if the components and parts that they buy from suppliers are of inferior quality or are steeped in unreliable supply, even though they may be priced competitively. This has the effect of creating a net cost disadvantage rather than an advantage. Experience has convinced many organizations in Japan that they need to develop a closer relationship with their suppliers, which should be characterised by a greater degree of trust and free exchange of information (both technical and commercial). Obviously, the more collaborative relationship cannot be maintained easily with several competing organizations because of the (i) costs and efforts involved in establishing such a relationship, and (ii) the costs and efforts of sustaining that relationship in the long run.

4.3 QUALITY-RELATED COSTS

Having briefly explained the benefits, we now turn our attention to quality costs or quality-related costs. Quality-related costs are costs incurred by an organization to ensure that the products or services it provides conform to customer requirements. In other words, quality costs are the sum of money spent on ensuring that customer requirements are met and also the costs wasted through failing to achieve the desired level of quality. Quality cost is thus the cost of not meeting the customers' requirement i.e. the cost of doing things wrong. In quality management, quality cost equals actual cost minus no failure cost. That is, the cost of quality is the difference between the actual costs of making and selling products and services and the cost if there were no failures during manufacture or use and no possibility of failure.

TQM philosophy adherents have defined quality costs as costs incurred in excess of those that would have been incurred if the product was built or service performed exactly right the first time. Costs would include not only those that are direct, but also those that result from lost customers, lost market share and the many hidden costs and foregone opportunities not identified by the cost accounting system.

Faigenbaum is attributed in developing the quality cost measurement while working in the General Electric Organization in the 1950s. He proposed a system of Quality Cost Reporting whereby all relevant costs were categorised into four major areas. These are:

- **Prevention costs**
- **Appraisal costs**
- **Internal failure costs**
- **External failure costs**

Prevention costs relate to efforts to prevent failures. These are the costs that are incurred on preventing a quality problem from arising. The cost of planning, introduction and maintenance of a quality system, and costs of any action taken to investigate, prevent or reduce non-conformities or defects would be included in this category. It is expected, and reasonably so, that expenditure incurred on prevention would reduce all other costs.

Appraisal costs relate to testing, execution and examination to assess whether specified quality is being maintained. These are the costs that are incurred in assessing that the products or service conform to the requirements. This category would include costs incurred on testing of raw materials, parts and components and other goods received on purchase from outsiders, and all inspection and testing during production. Appraisal costs can be contained or reduced when there is emphasis on quality improvement.

Internal failure costs (or losses) arise due to internal failures. Such costs result from a product or service failing to meet the quality requirement prior to delivery. These costs arise within the organization because of the failure to achieve the specified quality, before transfer of ownership to the customer. Such costs are incurred because something was not done right first time. In theory such costs would disappear if there were no internal defects such as scrap, or there was no need for rectification or redesign, or there were no delays in the production process due to non-conformance and non-conformance generated shortages. Internal failure costs are the easiest ones to identify and examine for quality cost reduction.

Box 4.1 Overreaction to Inspection Failures

Inspection of parts, components or products can never be 100% fool proof. At noted times they may escape inspection or there may be inspection failures. It should be that any over-reaction to it maybe counterproductive. This is similar to over-reaction of production management to an inventory stock out. Typically a defective item

that has escaped detection by inspection leads to a tightening of all process control, additional test procedures and more intensive final inspection. Most of the time these added costs are appeasements rather than a hard-nosed, professional quality decision. It is neither appropriate to look for scapegoats nor to add unnecessary costs.

External failure costs arise from the rejection of the products or services by the customers due to poor quality. These are the costs that arise outside the organization, i.e. after the transfer of ownership to the customer, because of failure to achieve the quality specified. Such costs would not appear if there were no external defects, warranty claims, replacement costs, etc. While estimating such costs, one must be a little cautious as quantification may not reflect the full story. There are losses such as customer goodwill or future loss of sales which are difficult to estimate and can easily escape from being quantified (the latter type of costs are also known as indirect costs). External failure cost can be reduced by more rigorous checking inspection testing, which might increase internal failure costs as more failures are detected in-house.

Recognition of the relative importance of external failure costs has caused progressive organizations to broaden their perspective from product quality to total customer satisfaction as the key quality goal.

Box 4.2 lists the source of items of costs that fall in each category.

Box 4.2: Various Types of Quality Costs

Prevention Costs

- Design/Review
- Quality and reliability training
- Quality planning including vendor quality planning
- Design, development and installation of quality measurement and test equipment
- Calibration and maintenance of quality measurement and test equipments
- Quality auditing
- Acquisition analysis and reporting of quality data
- Quality improvement programmes and quality engineering (preparation of quality manual, quality plans, standards, and procedures)

Appraisal Costs

- Test and inspection (or lab acceptance testing, installation testing, also installation and commissioning in high-tech environment)
- Maintenance and calibration
- Test equipment depreciation

- Analysis and reporting of tests and inspection results
- Line quality engineering
- Vendor rejects

Internal failure costs

- Faulty goods or scrap/rejects (as a result of tests and inspection)
- Rework and repair
- Design changes
- Trouble-shooting or defect failure analysis
- Re-inspection and retesting
- Downgrading
- Downtime

External failure costs

- Complaints
- Dealers' customers' rejects returns
- Commissioning failures
- Warranty claims
- Product reliability compensation (voluntary or legal)
- Concessions
- Loss of sales
- Recall costs

Prevention of costs, therefore, should receive greater attention at the hands of management because it is much less costly to prevent a defect than to correct it. The principle is similar to the traditional medical axiom: "An ounce of prevention is worth a pound of cure". The relationship between these costs is interesting to observe and is reflected in 1- 10-100 rule. One rupee spent on prevention will save 10 on appraisal, and 100 on failure costs. As one moves along the stream of events from design to delivery, the costs of error escalates and failure costs become higher and the pay off from an investment in prevention becomes greater. Computer system analysts are aware of this and understand that an hour spent on better programming or design can save up to ten hours of system retrofit and redesign.

Of all the costs explained above, only prevention costs are thus justifiable elements of the quality costs although appraisal may be justified in industries where safety is an important issue. Since, in theory the costs incurred in appraisal activities and failures are totally avoidable, they are jointly described as the cost of poor quality (COPQ) which, typically, may range between 15-40 per cent in industry. You may ask a question: if COPQ is that important or

significant, why do organizations not take any action to reduce it. The answer is that many organizations fail to identify the causes or drivers of such costs and even if some organizations do identify such costs, they consider them inevitable, e.g., over-ordering of material to allow for scrap, redrawing excess inventory levels, penalties for late deliveries, etc. If such costs are to be reduced then they must be identified and their causes known. Besides, they should also be reported to management. This calls for a method of classification and collection, or a system of quality costing and reporting.

A distinction must be made between External Failure Costs and External Assurance Quality Costs. The latter are costs that relate to demonstration and proof required as objective evidence by a customer, including particular and additional quality assurance provisions, procedures, data, administration tests and assumptions (e.g. the cost of testing for specific safety characteristics by recognised independent testing bodies). Such a practice is common with high technology products, using computer-based emulations and simulations to mimic the real world accurately in the test environment. Independent testing is considered vital for establishing the credentials of any product which, if it failed, could have product liability implications for its producer. External assurance costs would also include cost incurred by an organization for engaging an independent assessment organization to perform a detailed (or in-depth) study of all areas of an organization's activity which could affect the quality of its products or services.

A distinction should also be made between capital costs and revenue or recurring costs of quality. Unless otherwise stated, we are more concerned here with revenue or recurring costs of quality. While the first two types of costs i.e. prevention costs and appraisal costs are known as operating quality costs as they are incurred by a business to attain or ensure specified quality levels. (This would also include external assurance quality costs), the failure (internal and external) costs are known as non operating costs.

Direct and Indirect Costs

Quality costs may further be divided into Direct Costs and Indirect Costs. Prevention costs, appraisal costs, internal failure costs and some of the external failure costs are regarded as direct costs. While operating quality costs of prevention and appraisal are considered to be controllable quality costs, internal failure and external failure costs are regarded as uncontrollable.

You will note that as the controllable costs of prevention and appraisal increase, the uncontrollable costs of internal and external failure decrease. At some point, the cost of preventing and appraising defective products exceed the cost of correcting for product failure. This point represents the optimum operating quality cost.

Indirect (quality) costs can be divided into three categories:

1. **Customer-incurred quality costs:** These include costs to the customer due to equipment downtime, repairing costs after warranty period and transportation costs.

2. **Customer dissatisfaction costs:** Such costs arise because of customer dissatisfaction and as such are of attribute type. Such costs are high when defect levels are high and low when the defect levels are low.
3. **Loss of reputation costs:** These costs are related to the attitude of customers toward the organization. Because of lowering of an organization's image or esteem in the perception of the customers, future sales are adversely affected. Hence, such costs are the costs arising from foregone opportunities due to loss of reputation.

The indirect costs are mostly intangible and are difficult to measure. However, they do affect the quality cost curve. Prevention and appraisal costs will lower the external failure costs. A lower external failure cost has a desirable influence on indirect costs.

Though the measurement of actual indirect costs may be difficult, the concept is still valid. A knowledge that such costs do exist can be useful in economic analysis.

Magnitude of Quality Costs

The cost of quality or more specifically the costs of non-quality is something that should concern not only the individual organizations but also the national governments. The cost of poor quality may vary from organization to organization. However, in general, it is believed to be as high as 25% of sales. Some estimates of quality cost in American organizations indicate that they work out to 25% of revenues, whereas in Japan they are less than 5%.

4.4 LIFE CYCLE COSTS

While dealing with costs, it would be desirable to think in terms of life cycle costs, i.e. total costs (cost before and after the change of ownership). In other words, we should look at costs beyond the costs to the producers. Costs also arise outside the production organization once the ownership of the product or sales has been transferred to the user. Costs to the clients are related with acquiring, operating and maintaining the products, service. These would include adapting the product to client facilities, modification especially in critical systems. The client costs would also include idle time costs and revenue losses. Life cycle (quality) costs can be reduced by reliable maintainable and safe products manufactured by conforming to quality parameters. It is necessary to find optimum set of parameters which would minimize the total life cycle costs.

Life cycle costs may escalate where there is a lack of management commitment to quality. The attitude of lack of commitment may flow down through the organization to the shop floor. All products (or services) go through life cycle phases approximating to initial conception, feasibility, engineering system design, manufacturing/production, component and integration testing, system and client acceptance testing, operations and maintenance, and withdrawal from operations or service. As we move through this life cycle phases, the cost of correcting defects and failures increases dramatically.

You will see that if a defect is not detected at the initial incoming inspection stage, it may cost 1000 times more when the products in the field with the consumer/customer after the sale event.

Activity 1

With reference to your own organization and/or the industry/service you are in, estimate the cost of poor quality. Has the cost of poor quality been increasing or decreasing? (You may like to do some library research)

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4.8 QUALITY AND PRODUCTIVITY

When we talk about quality and productivity, some important questions arise. Are productivity and quality related? Are they two sides of the same coin? Can an organization have productivity as well as quality? The answer to all these questions is, yes. A growing body of evidence indicates that there is a positive correlation between productivity and quality. Despite this, there are still misconceptions and apprehensions in several minds that productivity and costs have to be sacrificed if quality is to be improved. From the market point of view, however, it has been clearly established that only when productivity and quality are considered together can competitiveness be enhanced. Those who believe that increased quality means decreased productivity argue that a programme to improve quality causes disruption and delays that result in reduced output. While in some cases it may be true in the short run, it is not so in the longer term. Such an argument fails if costs associated with poor quality are considered.

That there is a positive relationship between quality and productivity was strongly advocated by Deming. He forcefully argued that reduced productivity was caused by quality defects; rework and scrap. Feigenbaum maintained that a certain 'hidden' and non productive plant exists to rework and repair defects and returns, and if quality is improved, this hidden plant would be available for increased productivity. It will be seen that these arguments are straightforward. Any quality improvement that reduces defects is by definition an improvement in productivity. The cost of quality improvement rarely exceeds the savings from increased productivity.

However, it may be seen that building a case for or against quality improvement based on output or defect reduction alone may be oversimplifying the matter. A more convincing case can be built on basis of proven benefits of TQM. When the broader picture is considered it can be shown that increasing quality also increases productivity, and the two are mutually reinforcing. Productivity has come to mean more output for the same or less cost. TQM embraces a broader concept and can be considered to include the benefits of productivity

when properly implemented. Greater productivity can also be through attained cost reduction, greater efficiency, better use of resources and organizational restructuring, TQM goes a step further; it is a longer term and more comprehensive approach and is concerned with cultural change by creating vision, mission and values.

Role of Technology

The relationship between advanced technology and productivity has been a matter of great debate and discussion. Japanese success is mainly attributed to advanced technology. However, it is a mistake to consider technology to be the only answer to improvement in quality and productivity. It is not merely labour replacement which leads to more productivity, it is rather improvement in processes which is important and causes an upward thrust in productivity.

Many people think of technology as automation, mechanisation, machines and computers but the term has a much broader meaning. It is a means of transforming inputs into outputs. Technology thus includes methods, procedures, and techniques which enable this transformation. It includes both machines and methods. The latter improves processes which improve the output input ratio. Many organizations have achieved remarkable increases in both quality and productivity with little or no investment in the hardware side of technology. This is however not to discount the use of hardware to improve both quality and productivity.

However, automation and machines require capital investment, whereas management systems demand require that much capital and may be equally or more effective. The solution in several cases may lie in improving the systems and processes rather than in introducing new technology. Honda was able to market a more competitive car than General Motors by controlling cycle time and processes. The tendency is to focus on technology to reduce labour costs and to overlook the improvement in quality that can be achieved through improvement in related processes and tapping the potential of the workforce. Good organizations buy technology to improve proceeds, reduce leads time, boost quality and increase flexibility.

Experience has shown that frontline supervisors and employees have a wealth of innovative ideas for productivity and quality improvement. What an organization requires is proper systems; procedures and a right climate to harness their talent. The historical focus of cost reduction in the present context may not be adequate. In today's context capital outlay decisions must be based on overall productivity, improving the quality, cutting cycle time, redesigning inventory and adding flexibility. Activity analysis, and activity based costing (ABC) can be useful tools in improving management systems. We shall turn to these tools in a subsequent section.

4.6 APPROACHES TO QUALITY COST

Historically speaking there are two broad views on quality costs.

One view is that higher quality means higher costs. According to this view quality attributes such as performance and features cost more in terms of

material, design, etc. The additional benefits from improved quality do not compensate for additional expense’.

The other view is that the **cost of improving quality is less than the resulting savings**. Juran examined the economics of quality and concluded that benefits outweigh costs (though this view was originally promoted by Deming and is widely held by Japanese manufacturers). The savings result from less rework, scrap and other direct expenses relating to defects. This is said to account for increased focus on continuous process improvement in Japanese organizations.

Feigenbaum introduced “total quality control” and developed the principle that quality is everybody’s job, thus expanding the notion of quality costs beyond the manufacturing function. In 1979 Crosby introduced the now popular concept that “quality is free”. This view is held by those who adhere to TQM philosophy. Costs include not only those that are direct, but also those resulting from lost customers, lost market share and the many hidden costs and opportunities not identified by modern cost accounting system. If the products are manufactured or services are performed exactly right the first time, many costs are avoided. Philip Crosby is of the opinion that Zero defects should be the absolute performance standard and the cost of quality is the price of non-conformance against that standard.

Approaches to Quality Costs

Intuitively, the argument that improvement in process and product quality comes without costs may not appeal to those who are conditioned to the conventional mode of thinking. Let us examine the question of quality costs from two perspectives/approaches. The first one is the-Benefit Analysis approach, and the second is Deming-Kaizen approach. Let us take up the first approach first.

Cost-Benefit Analysis Approach

Under this approach, any improvement in the quality of the product or process is supposed to involve some cost to the producer. For example, greater conformance to quality standards could be obtained through such devices as more inspection, better materials, more maintenance, training etc., each of which requires a financial outlay. On the other hand, improved conformance to standards could reduce certain costs associated with poor quality, for example, scrap, rework, repair under warranty, and the cost of litigation. According to Juran and Gyrna,” organizations taking the cost-benefit approach will invest in improved quality so long as the additional benefits (cost reduction due to improved conformance) are greater than the additional expenditure required to improve conformance.

The Deming-Kaizen Approach

Not agreeing with the Cost-Benefit approach, Deming and others argue that a downward shift in the cost occurs hand-in-hand with improved product quality. For example, when internally generated improvements in production result in a higher percentage of throughput meeting engineering specifications, scrap, rework, repair rates etc. will be lower and the requirements for in-process

inventories and inspection will be reduced as well. The crucial point to note here is that an organization believing in continuous improvement (Kaizen) of product and processes is able to achieve cost reduction or economies as part of the day to day operations. The organization has not to incur special expenditure on implementing improvements to achieve the economies.

Let us take an example of an intermediate product say paper which is used by most other industrial organizations for their finished products. From the buyer's (or customer's) point of view, quality will be conceived in terms of strength, consistency of finish, width of the roll, thickness of paper, total weight of the roll, etc. Any variations in any of the properties of paper, as spelt out in specifications, can become a source of problem for the buyer in the conversion process. With optimal paper (i.e. paper with optimal properties), the conversion time would be minimal. However, with variations in any of the properties attributes of the paper, actual time taken by the conversion process would be more. For example, if paper is thinner than what it should be, the speed of conversion line per force has to be slower to prevent tears in the paper.

In the event of the tear, the line must be stopped and the paper restrung. To prevent any unsatisfactory rolls of paper (with thinner, under weight, etc.) being received at his factory, the buyer has to establish some screening inspection process for incoming shipments and thus entail costs, for example, in terms of space, equipment, personnel, etc. As such s/he may refuse to accept/take rolls which underweight or have some other defect (i.e. variations from specifications). Sooner or later, the refusal decision would push the cost of inspection back to the producer of paper rolls. Thus, the refusal to accept paper has a cost implication for the producer/seller.

The question of variations, you must note, is a very important one from the quality of point. Variations on both sides of the spectrum (e.g. underweight vs. overweight, thinner vs. thicker) beyond the acceptable level means rejection which in turn means higher costs (scrap, rework, higher transportation costs, higher inventory, etc.) for the supplier and the buyer. A reduction in variations is tantamount to improvement in quality as it eliminates some of the costs associated with poor quality.

In a Kaizen organization, the employees/workers work under a broader framework. They are rotated quite frequently across a number of jobs. They are promoted across departments and hence have a more comprehensive perspective. Apart from being more productive, they have an in-depth understanding of Work product processes that encompass many functional specialities, from start to finish. Individual members of the organization would understand how each speciality fits into the bigger picture. An employee worker is an informed generalist specialising in a number of jobs and can communicate across functions unlike the isolated specialist in a traditional organization. The gains from improvement from this approach are likely to outweigh the sacrifices of economics of specialisation." A Kaizen organization organizes and manages itself in such a way that improvements happen in the normal day-to-day operations.

4.7 QUALITY COSTING AND COST SYSTEM

Every organization, more so a business organization (whether manufacturing or service), should be aware of the cost of quality. The senior management must see to it that quality costs are identified and measured as part of the organization- wide policy, failing which the organization may fail to perform competitively. Increasing consumer protectionism has been reinforcing social and individual pressures to meet the requirements of quality. An organization needs to establish a quality cost system which may be limited to the costs incurred by it as a manufacturer or supplier of product and services or it may extend to the users operational failure costs.

The latter approach (life cycle cost approach) attempts to collect costs according to the whole life cycle of the product or service and gives total cost of quality across all activities of the organization and its clients. Only when management knows the total cost of quality can it have an appreciation that the cost of improving quality works out to be much smaller compared to the cost of not having a quality cost system. With this realisation it may be better that the allocation of more resources and better trained personnel should be kept for quality related activities. It is not only desirable but rather necessary that quality costs are regularly monitored and reported to the management for appropriate decisions and actions.

Although the need for quality costing cannot be overemphasised it will not solve quality problems by itself. It must be used as part of a wider total quality management approach, where problem areas are identified and prioritised, using quality cost data and initiating improvement activity. Quality costs are usually incurred across departmental lines and involve all activities of the organization -purchasing, manufacturing, design, and service, to name a few. Some costs such as inspection, instrumentation and quality personnel salaries are readily identifiable. Other costs, such as those associated with scrap and rework, are not that easy to identify and allocate. There are costs related to customer dissatisfaction and loss of reputation which are difficult, if not impossible, to measure.

Only by knowing the points (in terms of time or place) where costs are incurred can management know the magnitude of such costs and monitor and control them. Quality costs must be collected and recorded separately otherwise they get absorbed and concealed in numerous overheads: Many organizations attempt to measure quality costs and compile the information and statistics outside their accounting system. An effective cost of quality planning and control system should be directed towards the basic reason for quality failure improvement. Activities and functions are not independent. They form a system of interdependencies that are connected by linkages and relationships. For example, purchasing from a low quality supplier may lead to redesign, rework, scrap, etc. These linkages are difficult to recognise and are often overlooked. The conventional accounting system is generally not equipped to separate the cost of quality in these linked activities. Virtually all accounting classifications and activities along functional lines and the reporting of quality costs is done into several general expense, categories such as salaries, training costs, etc.

As one of the steps in the design of planning and control system it is useful to identify those activities and linkages between activities where costs occur. Some form of linear or matrix organization chart or table could be useful for this purpose. Departments or activities are listed across the top and costs of quality down the left-hand side. Regular financial reports to management are vital and impart visibility to quality related costs. Quality costs can be related to other costs or measures (e.g. ratios can be calculated) such as sales turnover, added value, etc.: (i) to evaluate the adequacy and effectiveness of quality management system; (ii) to identify additional areas requiring attention, and (iii) to establish quality and cost objectives.

Budgeting, Control and Reporting of Quality Costs

Like any other activity, quality costs should be budgeted and controlled. Mechanisms of control should be established. Actual costs on various items falling in various categories should be compared with budget amounts and variances computed. Only when regular financial reports are available can management be in a position to identify critical quality-related cost areas. A Pareto analysis of the vital few from trial among quality-related cost areas will enable management to rationally allocate its resources and decide priorities.

Some Useful Techniques for Budgeting, Control and Reporting

Some of the tools and techniques which can be helpful in budgeting control and reporting of quality costs are discussed here under.

Operating Quality Cost Report: The basic quality cost instrument is the operating cost report which is usually issued by the accounting department. Provision is made in the report for monthly cost information and year-to-date cost information by cost category. A column is included for reporting variances from budgeted or planned quality costs. There is a tendency to analyse only unfavourable variances. This practice can be a mistake since the analysis of substantial favourable variances may provide information leading to formulation of programmes that would continue this happy state of affairs.

Trend Analysis: Trend analysis can provide information for long range planning, particularly where monthly quality cost reports are prepared which are essentially based on short range perspective. Trend analysis can be done based on cost category or sub-category, by product, by measurement base, by plant within a organization or by a work centre. Trend analysis is an effective tool provided it is recognised that some period-to-period fluctuations may be only normal variations.

Ratios are the most commonly used management tool for monitoring and evaluating performance. Some useful ratios for evaluating the efficacy of the quality programme. An organization must decide on what ratios would be useful for it. Care should also be exercised in interpreting ratios.

Zero Based Budgeting: The concept of Zero Based Budgeting (ZBB) which provides an analytical framework for evaluating projects and activities in terms of overall objectives, goals and policies can be gainfully used for quality cost

control. ZBB requires the involvement of all levels in the budgeting process and provides a credible rationale for allocating/reallocating resources.

Pareto Analysis: Pareto distribution is an effective tool for cost analysis. The crux of the Pareto principle is “concentrate on few” as it increases cost effectiveness. To apply the Pareto principle, management must first identify the potential quality problems or problem areas, and then identify the vital few problems or defects e.g. scarce material, critical items requiring extensive quality assurance, high-tech parts, large capital equipment projects, computer software sub-contracts, items recently switched from a make to a buy, etc. Pareto principle may be applied in physical or quantity (number) terms, or it may be applied in financial terms. It may be desirable that losses are translated into financial terms, and non-recurring costs and time factor are considered. Only then the clear picture would emerge.

The past can guide an organization in preparing the Pareto distribution. The items may be analysed in order of frequency of occurrence and identified by letters such as A, B, C etc. It may be found, for example, that only three defects, viz. A, B, C are responsible for 80% of all defects that occur. These defects are the vital few. The question is: should the organization attack these three defects A, B, and C? The answer may be yes, but only if the organization's objective is to reduce the total number of defects (or the average number of defects per unit).

Attacking defects A, B, and C may not really help the organization if they have a minor impact on quality in monetary terms. It may therefore make more sense for the organization to assign a monetary value to each of these defects to make a truer picture of the cost of defects. On doing this, for example, it may be found that some other defects, say E, G and K account for nearly 80% of quality defects in monetary terms. Therefore, it may be more desirable to concentrate on defects E, G and K. Further, the nature of some defects, say defect G for example, may be such that it may take several years to remove the defect. For example, a machining operation may have to be, critically redesigned. Which defect should receive more attention of the management would also depend upon the latter's time orientation.

Activity Based Costing (ABC) : Activity Based Costing attempts to collect information on financial and operational performance and traces the significant activities of a organization to process, product and quality costs. ABC can prove to be a highly suitable technique for TQM because it encourages management to analyse activities and determine their value to the customer.

It may be said that costs are not incurred or allocated, they are caused. The mere collection of data is of little use unless the data can help identify the drivers of quality costs so that problem identification can lead to problem solution. It is in this context that ABC can be extremely useful and helpful. If the cost driver relates to an activity which does not add value, it can be eliminated or reduced. One can do this by examining the cause and effect relationship. Cost drivers are the agents that create or drive the need for an activity and hence cause resources to be consumed. It is estimated that 50% or more of the activities in most businesses are cost added rather than value added.

The basic concept of ABC is that costs of products and quality should be traced to the drivers of activities that consume resources and cause these costs. Many organizations often fail to compile the many costs relating prevention, appraisal, internal failure, and external failure that are hidden until identified by a quality cost (management) system. If the costs are not identified there is little chance of tracing them to the processes or activities that are causing them. Only the “visible” rework, scrap and repair/service costs are compiled by most of the organizations, whereas all costs, visible and non-visible should be compiled or assessed.

Value Analysis Value Engineering: Value analysis value engineering can be useful techniques in quality cost control. Value analysis involves examining a product or operation to eliminate any cost elements that are not essential to the effectiveness, utility or marketability of a product or operation. In the case of a product, this involves simplifying the design, substituting with lower cost material, improving reliability or durability (particularly where a organization is subject to warranty costs) or eliminating unneeded parts.

The purpose of value analysis is to provide the required functions at a minimum cost, without sacrificing the required performance or quality (of the product/service) i.e. without affecting the value of the product or service to the customer. When applied to quality control operations, value analysis can provide ways to eliminate unneeded or expensive steps that do not contribute worthwhile benefits. -

4.8 USES OF QUALITY COST INFORMATION

The potential uses of the information contained in quality cost reports are numerous but may be grouped into three broad categories. First, quality cost may be used to promote quality as business parameter. Second, it may also serve the motivational purpose. Third, it provides the means for planning and controlling future quality.

Quality as a Business Parameter: Information of this type aims at gaining the attention of higher management by using the language of finance. This allows quality to be treated as a business parameter like any other activity. Quality cost information giving rise to performance measures is usually expressed as absolute or relative costs i.e. indexes or ratios to other business costs. These cost measures may be used for these purposes:

(a) comparison with other parts of business or with other businesses; (b) decision-making.

The current performance may be compared against past performance. It is important that both these sets of cost data should be arrived at on the same basis. These apart, quality costs can also be used to learn more about the economics of investment in prevention activities. The use of quality costs for decision-making refers to its use in helping managers to identify major opportunities for cost reduction.

Quality for Motivational purpose: The use of quality costs for motivational purposes include displays to shop floor workers of scrap costs arising within

their centres and to middle managers for their department's contribution to total quality costs.

Quality costs for planning and controlling: The use of quality cost as a means of planning and controlling in dudu establishing quality cost budgets for the purpose of cost control.

More specifically quality cost information can be used in a number of ways as follows:

- To identify profit opportunities (every penny saved goes to the bottom line)
- To make capital budgeting and other investment decisions (quality, as opposed to payback, is the driver of decisions to purchase new equipment or dispose of unneeded ones; equipment for rework is not needed if the rework is eliminated or reduced)
- To improve purchasing and supplier-related costs
- To identify waste in overhead caused by activities not required by the customer
- To identify redundant systems
- To determine whether quality costs are properly distributed
- To establish goals for budgets and profit planning
- To identify quality problems
- As a management tool for comparative measures of input-output relationships (e.g., the cost of a reliability effort versus warranty costs)
- As a tool of Pareto analysis to distinguish between the “vital few” and the “trivial many”
- As a strategic management tool to allocate resources for strategy formulation and implementation
- As an objective performance appraisal measure

Some organizations e.g. General Electric's quality cost system emphasise non-product features such as inquiry responsiveness, delivery times and billing accuracy. The emphasis is on root cause analysis and process improvement: simplification of procedures and reducing cycle time and pushing down in quality costs while improving customer satisfaction.

Activity 2

Design a quality cost reporting format suitable for use in an organization of your choice. Discuss the information, collection and analysis requirements with your management accountant and prepare an initial quality cost report.

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4.9 SUMMARY

Economics of quality i.e. costs and benefits of quality are an important subject in TQM. If an organization enjoys a reputation in the market for quality products, this in itself becomes a source of competitive advantage. Besides, benefits like increased market share, greater customer loyalty, higher share prices, greater productivity, enhanced employee morale, improved productivity etc. are some other tangible benefits that are likely to occur to the organization.

The quality-related costs can be classified into four categories: Prevention costs, Appraisal costs, Internal Failure costs, External Failure costs. Prevention costs should get the most emphasis as it is much less costlier to prevent a defect than to correct it.

Quality costs can be further divided into direct costs and indirect costs. While prevention and appraisal costs are considered controllable, failure costs, whether internal or external, are regarded as uncontrollable. Life cycle costs take into account all the costs from cradle to grave i.e. through the entire product life cycle.

That there is a strong relationship between quality and productivity has been stressed by quality gurus, particularly by Deming and Feigenbaum. A quality improvement programme that reduces defects in itself is an improvement in productivity. Technology has an important role in raising productivity. In its broader interpretation, and being concerned with transformation of inputs into outputs, technology includes both machines and methods. The latter is the softer side of technology, and many organizations have achieved remarkable increase in both quality and productivity by strengthening improving methods and processes rather than investing heavily in hardware. Broadly speaking, there are two views on quality costs. The traditional view is that quality means higher costs. The other view is that benefits of or saving from quality are much higher than the cost.

According to cost benefit approach, organizations will continue to invest in improvement in quality so long as additional benefits are greater than additional expenditure required to improve conformance. Contrary to this approach the Deming-Kaizen approach postulates that a downward shift in cost of quality occurs hand in hand with improved product quality.

For realising full benefits of quality policy and programme, an organization needs to have a quality cost system whereby quality costs are systematically identified and measured. Further, an organization needs to have a budgetary system on a continuous basis for computation of variances to trigger appropriate reporting and action for future. Zero Based Budgeting, Ratio Analysis, Value Analysis and Engineering, Activity Based Costing, Quality Cost Reports, Trend Analysis, and Pareto Analysis can be useful devices in this context.

Quality cost information can have several uses. It can be used: as a business parameter for decision-making purposes, for planning and controlling costs, and for motivational purposes.

4.10 KEY WORDS

Quality Engineering costs	: Cost of prevention activities associated with the creation of overall quality plan (i.e. implementation, maintenance and auditing inspection plan, reliability plan, data system and other special plans).
Design and development of equipment	: Cost of personnel engaged in planning and measurement of quality control equipment. One of the prevention costs.
Appraisal cost or Product	: Expenses involved in performing quality.
Quality Audit	: Audits on in-process or finished products.
Scrap	: An Internal Failure cost equal to net loss of labour, material and overheads resulting from economically non-saleable/non-reparable defective products.
Rework	: Cost of correcting defective units so that they meet specifications.
Failure Analysis	: Costs incurred to determine the cause of product failure.
Rejected and returned	: External failures costs associated with handling and replacement of returned products.
Quality Assurance	: It embraces all activities and functions concerned with attainment of quality.
Quality System	: A system that embodied the structure, responsibility, procedures, processes and resources for implementing quality management
Prevention Costs	: Costs incurred to prevent failure (internal and external) i.e. costs incurred in preventing quality problem from arising.
Appraisal Costs	: Cost incurred on testing, execution and examination to assess whether specified quality is being maintained.
Internal Failure Costs	: Cost arising from internal failure i.e. products/ services failing to meet quality requirements prior to delivery.
External Failure Costs	: Costs arising from rejection of products or services by customers due to poor quality occurring after delivery.

External Assurance Costs	: Costs relating to demonstration and proof required as objective evidence by customer, particularly in the case of high tech, high risk or critical products.
Direct Quality Costs	: Costs directly measurable in monetary terms arising from poor quality of products or services. Prevention, appraisal, internal failure and some of the external failure costs are direct quality costs and can be measured without much difficulty.
Indirect Quality Costs	: Costs not directly measurable in monetary terms due quality being below the expected levels. Such costs include customer incurred costs; customer dissatisfaction cost and cost arising from loss of reputation. These are difficult to measure.
Life Cycle Costs	: Costs of non-quality throughout the product life cycle i.e. cost incurred before and after change of ownership.
Cost-Benefit Approach	: A purely economically oriented approach under which a organization is supposed to continue to invest in improving quality as long as the additional benefits from improved conformance are greater than additional expenditure.
Quality Cost System	: A system, in which costs of quality (or non quality) are identified, measured and reported on a regular and systematic basis.
Activity Based Costing	: A method of costing that traces the costs to the cost drivers (or activities) that cause them.

4.11 SELF-ASSESSMENT QUESTIONS

1. What benefits is a quality conscious organization likely to derive from effectively implementing a quality programme?
2. Discuss the various types of quality costs. What kind of cost should an organization be more concerned with?
3. Why should prevention cost take precedence over the other three classifications?
4. Examine the statement that “Quality is Free”. Give reasons in support of your answer.
5. Discuss the meaning and significance of life cycle costs in connection with quality.

6. Are quality and productivity related with each other? Discuss fully, citing any examples of business organizations which have been able to improve both quality and productivity.
7. Discuss and examine different views on costs and benefits of quality. Which of the views, do you think, you can support and why?
8. Why should an organization have a quality cost system? What uses has the quality cost information for the management of a business organization?

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UNIT 5 TQM AND BUSINESS STRATEGY

Objectives

After reading this unit you should be able to understand:

- the relationship between Total Quality and business strategy;
- the meaning of customer value strategy, and its significance, and the need for customer value determination systems;
- the shift in thinking about buyers and suppliers required in TQM;
- the elements of quality business plan; and
- the reasons behind success and failure of quality strategy programmes.

Structure

- 5.1 Introduction
- 5.2 TQM and Corporate Strategic Process
- 5.3 Total Quality and Customer Value Strategy
- 5.4 Total Quality, Cost Leadership and Differentiation
- 5.5 Customer Value
- 5.6 Customer Value Determination Systems
- 5.7 TQM and Stakeholders
- 5.8 Total Quality and Corporate Strategic Alternatives
- 5.9 Quality Business Plan
- 5.10 Success/failure of Quality Strategy/Programmes
- 5.11 Summary
- 5.12 Key Words
- 5.13 Self-Assessment Questions
- 5.14 Further Readings

5.1 INTRODUCTION

Total Quality Management (TQM) is a systems approach to management with the aim of continuously improving the value to customers by designing and continuously improving organizational processes and systems. A variety of external forces, including the policy of liberalization pursued by many countries of the world and establishment of WTO with consequent reduction in trade barriers, increased foreign competition, increasing demands by worldwide customers have paved the way for quality management. These forces have made the organizations realize the potential of competitiveness provided by quality. Organizations producing consistently high quality products and services offer them at competitive

prices have a strong advantage. They can do better at the competitive market place. Organizations which can offer high quality products at reasonable prices have the final word on how well a product fulfils the need and expectations of the customers/end users. The satisfaction about a product is always related to competitive offerings. The relatively deregulated global competition has enabled choice for the customers in many industries.

For example, couple of decades ago the only relevant choice that a customer of automobile within the country had was limited between a few brands but now so many passenger cars of Indian and foreign makes are available. The choice of automobiles is available to customers in a wide range of products and services. The choices, as a matter of fact, seem to have no end. In such an environment, customers demand high quality, new technology, low price, among many other things. The international competition has taught the organizations in many countries that they must deliver superior value to customers in order to retain them. Therefore, quality management as a part of business strategy for the organizations has become all the more important. The importance of quality in organizations started way back in the 1990s.

In this unit we will learn several aspects of strategic management of quality. We will discuss the relationship between TQM and Corporate Strategic Process, Customer Value Strategy, TQM and the new thinking about buyers and suppliers, Total Quality and Corporate Strategic Alternatives, Quality Business Plan, etc.

5.2 TQM AND CORPORATE STRATEGIC PROCESS

You might be aware that corporate strategy consists of two key phases. The first phase is the strategic formulation which necessitates defining the mission of the organization, and this remains unchanged over a fairly long period of time. In this phase the strategic options are also defined and the optimum option is chosen. The result of this phase is the strategic plan. The second phase is the strategic implementation which is mainly concerned with operations management. It also defines the short-term plan for the organization. A question may be asked: Where does TQM fit into the corporate strategy? There may be different answers to this question. Most people think that TQM should be linked to operations management. This is by no means a coincidence because before the Japanese developed TQM, they started with industrial engineering, quality control, Organization Wide Quality Control (OWQC), and value engineering. Unfortunately, it is not always obvious that during the evolution of TQM, the concept of quality has filtered up the hierarchy. Today, the Japanese managers and directors are so concerned about quality that it has long become their mission. Consequently, in their strategic formulation process, they use quality as their key mission statement and as an integral element of strategy.

5.3 TOTAL QUALITY AND CUSTOMER VALUE STRATEGY

In the context of Total Quality, business strategy, as we have stressed earlier, should be seen as synonymous with customer value strategy. How should the organization provide value to customers in its chosen business? This question

warrants serious attention as it is associated with corporate-level strategy concerning the choice of business. It has been argued that to be successful today organizations should enter and invest only in businesses where they can be quality and value leaders.

Customer Satisfaction and Delight

Total Quality (TQ) as a business strategy focuses primarily on customers by consistently improving value to customers, and thereby delighting them. Unlike competitive strategy whose focus is on competitors, the focus under TQ (or customer value strategy) is on customer. The organization attempts to retain current customers and attracts new ones by delivering best customer value. It thus aims at growth in market shares. Customer is at the core of the definition of TQ and customer focus is one of the critical factors in TQ implementation in all types of activities, whether manufacturing or service, whether in public or private sectors. Big and successful organizations have understood the importance of customers for their success and describe total customer satisfaction as their foremost goal. They are committed to quality, value, and customer satisfaction.

Now the question arises as to what is customer satisfaction? Customer satisfaction is concerned with the customers' positive or negative feelings about the value received from an organization's products or services. Meeting customers' requirements and satisfying customers is one level of customer commitment. However, the point to be noted is that striving to satisfy customers by meeting their expectations implies a reactive mode to provide them what they expect of the organization. Total customer satisfaction and customer delight by exceeding customers' expectations is different from the earlier reactive mode. It is more proactive and goes beyond what customer demands today so as to keep him as customer tomorrow. It may cost an organization several times (may be four or five times) as much to attract a new customer than to retain an old one. Such a commitment may mean providing better value to customers than other competitors.

Competitive Strategy and Customer

Competitive strategy is an important component of strategic management and involves in-depth study of competitors and their strategies, their products and operations. Competitors are at the heart of Porter's generic competitive strategies and are at the center of Porter's Five Forces Model.

Competitors or Customers

It has to be appreciated that a preoccupation with the competitors should not be at the expense of concentrating on the customers needed under TQM. The best competitive strategy in today's environment is a customer value strategy. By concentrating on exceeding customers' expectations in the TQM mode the organization will be in a better position than its competitors to retain current customers and win new ones. If an organization concentrates on current products or services of its competitors, it may unwittingly result in the organization's pursuing a 'follower' mentality. Such followership may result in imitation rather than innovation. By concentrating principally on competitors, the organization may easily lose sight of customers. If an organization merely imitates the competition it cannot be an early market entrant with new product or service.

In the present scenario when the customers are constantly demanding new products or services with latest technology, late market entrants, by ignoring customers, have to pay a high penalty. This, however, does not mean that the organization should not benchmark competitors' best processes and systems. An organization should develop new products or services by a thorough examination and understanding of the market (and this includes competitors' products) as well as customers' future requirements. Understanding the competitors' current products may not necessarily be the key to the design of new products or services. Customers may be more important for design of new products while competitors may be more important for improving processes or system design.

Let us now understand how high quality and low cost can go together.

Quality and Low Cost

Historically, high quality has been associated with high cost. The cost of quality includes all those costs that are incurred due to producing poor quality products and services. Such a cost includes costs of scrap, rework, warranty repair, inspection and quality related maintenance, etc.

The cost of quality, expressed as a percentage of cost of goods sold, may be as high as 20% to 30%. Experience shows that organizations implementing TQM programmes have been able to reduce the cost of quality by 90% and even more. Through relentless drive to improve critical cross-functional processes, some organizations have been able to most eliminate cost of quality (i.e. they have reached zero defect stage). Such organizations have demonstrated that high quality and lower costs is a realisable goal. The organizations can provide more value to customers as their prime business level strategy.

Total quality attempts to closely link strategies and systems. The systems and processes require a primary focus on external customers, and cross-functional horizontal spans. Cross-functional systems are activities and resources that span at least two functions of design, production, marketing and finance. Such systems are associated with horizontal structures and flatter s with fewer levels of management, less direct supervision, cross-functional teams, and employee training.

Activity 1

In the context of the organization you are associated with, examine whether quality is an integral part of the strategic process or not. Justify.

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5.4 TOTAL QUALITY, COST LEADERSHIP AND DIFFERENTIATION

Contrary to achieving just one of the strategies implied in the Porter's generic Competitive Strategy model an organization can achieve both cost leadership and differentiation simultaneously. Both these strategies, in fact, may be required to achieve sustained competitive advantage in the industry. High quality, innovation, learning effects, and economies of scale have been mentioned in the literature as sources of the conditions for achieving both generic competitive strategies.

Mintzberg has offered an alternative to Porter's three generic competitive strategies which include "differentiation by quality" as a generic strategy. Some other generic strategies included are: image, design, price and support. Several writers have contended that flawless quality and the systems that produce it are the principal ways to simultaneously achieve both low costs and differentiation. Cost leadership is associated with low cost of quality, high work force productivity, flat organization in structure, and short response times.

Differentiation is associated with a dedication to providing flawless quality and value to customers. Many Japanese organizations in several industries, including consumer electronics and automobiles, have successfully employed quality strategy. The list continues to grow.

5.5 CUSTOMER VALUE

When we say that the primary objective of quality is to give value to customers, it does not simply mean elimination of defects. Customer value has many dimensions which must be systematically reflected in the organization's product and services. Customer value determination and new product development are in fact the two most important systems of a business organization. A strong linkage between the two systems will enable the delivery of superior value to customers and sustained competitive advantage to the organization.

Quality Gurus have shown the business world that it could deliver high quality and low price to the customer simultaneously, as we have examined earlier. Quality just does not mean that the product has very few defects or no defects or that the product adheres to internal engineering specifications. Quality means diverging value to customers as per their expectations.

Quality is thus multi-dimensional. Garvin has listed eight dimensions of quality as under:

- Performance
- Features
- Reliability
- Conformance
- Durability
- Serviceability

- Aesthetics
- Perceived Quality

Schonberger has added the following four more dimensions to the list making a total of twelve:

1. Quick Response
2. Quick-change response
3. Humanity
4. Value

Realized Value, Sacrificed Value and Net Value

Value has two aspects: one aspect relates to value realized, and the other relates to value sacrificed. **Realized Value** is the value that the customer receives which may include comfort, image, ease of use, reliability, consistency, enjoyment, and a host of other characteristics. All these, in fact, relates to the various dimensions of quality. **Sacrificed value** is the value that customer gives up. It may include money, time, energy, worry, frustration, etc. Customer value is the customer perception of what has happened to her/him in relation to the product or service obtained in order to accomplish a desired goal. The difference between value realized and value sacrificed is the net value gained or lost. Net value may thus be either positive or negative. If the value realized is more than the value sacrificed, the net value will be positive and the customer will stand to gain. This may be reflected in repeated purchases. While a satisfied customer may tell other potential customers' and share her/his satisfaction with them, a dissatisfied customer may do just the opposite. It is, therefore important for managers to pay attention to the value sacrificed by customers as it could lead to dissatisfaction.

5.6 CUSTOMER VALUE DETERMINATION SYSTEMS

If providing best value to the customer is the quality objective of an organization, then the organization must pay particular attention to determining customer requirements. Managers in several functions must fully grasp the concept of value realized and value sacrificed. A workable cross-functional system needs to be formalized. Data has to be collected through large scale surveys and interviews with customers (present and potential) for determining customer requirements and for measuring satisfaction (present customers). Similarly in-depth interviews have to be conducted with small number of customers (or focused groups) for exploring problems. Sometimes customers are impressed just because the organization has cared enough to ask if they were satisfied. The managers may be held accountable on the basis of customer satisfaction data. The data helps to continuously improve the product and the manufacturing processes to yield even greater value to customers.

Concurrent product development by cross-functional teams, with which sometimes even large customers are associated, is a useful method of designing and developing products and services that would satisfy customers, another useful

method is QFD (Quality Function Deployment) which is meant to ensure that the design of new product or service is based on customer criteria. QFD starts with customer criteria, translates them into product or service requirements, and then translates them into product or service requirement measures.

It is customer driven design system that attempts to get early coupling between the requirements of customer, marketing, and design engineering. For example, in the design of a personal computer (PC), customer requirements might be expressed as ease of use, portability, quality of graphics, affordability, and speed. The product requirements might then be expressed as menu-driven commands, open architecture, compact size, high resolution screen, medium price, and fast micro chip. The product requirement measures might then be expressed as lightweight (say less than 4.5 kg), compact size (say 50 x 40 cms), and medium price (say 30,000-60,000).

5.7 TQM AND STAKEHOLDERS

Let us now consider how TQM affects the perception about some stakeholders, and how it affects the owners.

Buyers or Customer

The customer today should not be seen as a mere buyer, a term which connotes adversarial exchange. Porter's Five Forces model of industrial economics perhaps implies on adversarial exchange based on the bargaining power ideas. It suggests that if there is some way to weaken the bargaining power of buyer, then all is well and the organization need not worry about exceeding or even satisfying customer requirements.

Suppliers or Partners

In today's competitive environment which demands zero-defect production and delivery, many organizations have recognized the importance of suppliers and regard them as partners in their systems. The idea of supplier power envisaged or implied in Five Forces model again connotes adversarial mentality. Japanese organizations have excelled in forging partnership with their vendors/suppliers and have set an example for others to follow. A commitment for continuous quality improvement cannot be translated into reality without treating the suppliers as partners.

Supplier Partnering

Japanese concept of *Keirestu* has led many organizations to rethink about their relationships with suppliers. The central point in *Keirestu* is developing long term relationships with a few key suppliers rather than having short term relationship with many suppliers. Long term relationship is important since it has been widely recognized that suppliers have a profound impact on the value which is finally delivered to the customers.

The prevailing system of low bid contracting, under which contract is awarded for the year encouraged suppliers to cut down on quality. This system of contracting resulted in large number of suppliers, with high turnover of supplier relationships. Many organizations today find virtue in developing long term partnering relationships with suppliers and work with them for long term cost reduction and quality

enhancement. The partnering may even include training of the supplier's staff by the customer organization, especially in improving quality management systems. Several organizations have dramatically cut down on the numbers of suppliers (some have even cut down by 90%). Based upon a relationship of trust and long term mutual interest, many customer organizations are now willing to invest in a few key suppliers rather than requiring several suppliers to constantly bid against each other. The yearly low-bid contracting system requires the customer organization to inspect incoming shipments from the supplier for defects (quality being the concern of the buyer). Acceptance sampling has to be used to make a decision to accept or reject a shipment based on the number of defects found in an inspected sample.

Today, the customer organizations believe in the certification of the operational processes and systems of the supplier. Once the supplier's processes have been certified, the customer can eliminate incoming inspection. This facilitates just-in-time (JIT) system in which the incoming material, parts, etc. get immediately integrated into the customer operations. In a JIT inventory system (Japanese *Kanban* or Zero inventory system); materials arrive at the point of use exactly when they are needed. The inventory level practically reaches zero level, and the inventory holding costs approach zero. Obviously, a JIT system would work only where processes are in control, managers know exactly when to order, and the orders arrive as scheduled without any variance. Such a finely-tuned system, like clockwork, can work only when all parts are coordinated. If one part of the system creates a variance, the whole system may shut down quickly. JIT system may cover not only external suppliers but also internal suppliers and customers. It is a demand pull system because each station or worker produces the output only when the next work station or worker up the line is ready to receive more input. This is in contrast to the push inventory system in which parts and components are made irrespective of whether the next work station is ready to use them.

Does the primacy given to customers come in conflict with the interest of the owners? Does the focus on customers stand in the way of maximization the wealth of the shareholders, usually regarded as the objective of financial management? A growing body of evidence shows that TQM, with its focus on customers, is the way in the present day globally competitive market place to produce superior long term financial value for owners. For organizations with more advanced TQM systems, the improvement in financial performance, is notably and consistently stronger.

5.8 TOTAL QUALITY AND CORPORATE STRATEGIC ALTERNATIVES

TQM seems to go well with certain strategic options/alternatives. Certain factors which have emerged on the horizon over the recent past combined with TQM have influenced management to follow specific strategic courses and abandon some others. Let us examine this aspect.

Related Versus Unrelated Diversification

In a widely diversified organization, it is difficult, if not impossible, to provide superior value to customers and to continually improve that value in each of the different markets. It is therefore being increasingly felt that corporate strategies that emphasize concentration or related diversification are more likely to succeed

in the era of global competition and TQM. In contrast, conglomerate or unrelated diversification is likely to be de-emphasized in such an environment.

Sustained competitive advantage is more likely if the organization is committed to a particular kind of competitive strategy throughout its operations. Organizations are more likely to succeed in businesses where they have the chance of becoming quality and value leaders since such leadership only can provide sustained competitive advantage. An example of organizations having sustained competitive advantage is the Wal-Mart corporation. The concept of low-cost operations with low cost and low prices, which permeates throughout the organization, has given it a sustained competitive advantage. Wal-Mart is regarded as the world's largest discount retail store.

Activity 2

Describe :

- (i) how an organization defines value to the customer?

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- (ii) how it determines the value being provided to the customer. Does it have any formally laid down process?

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- (iii) how it views various stakeholders in the context of TQM?

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5.9 QUALITY BUSINESS PLAN

Quality has emerged as an integral part of doing business these days. Quality and business, therefore, are one and indivisible. Quality will not happen just because the top management states that it has to happen. There has to be an effective business plan to provide clear view of the road ahead. The quality business plan will define the destination for its members, the strategies and the organization need to complete the journey. The business plan must establish a sound and stable base to ensure that the journey will be kept on contemplated course. It should provide organization guidelines for those who are responsible for implementation of the quality plan. This is essential for building the climate of commitment and communication.

A well established organization for quality is *sine quo* none for implementing the quality plan. The organization for quality should fit with the structure and nature of the business. The organization may include a steering committee, a number of teams (TQM teams) and task forces to be decided by the chief executive officer (CEO), with support provided by TQM coordinator and the external consultant. One of the members of the team may become the quality champion for a particular unit or function of the organization. The quality champions are facilitators and assist the teams to implement the process of change. The facilitators, who are provided specialized education at the commencement of the planning phase, have to work closely with the coordinator and consultant to develop the detailed plan for submission to the Quality Steering Committee.

The quality business plan may be developed, in association with the external consultant, if necessary, and may include the following elements:

- Quality mission of the organization
- Quality strategy, policies to be adopted by the organization
- Principles and values to be shared by the whole organization
- Organization and the processes
- Education plan to involve all employees
- Tactical plans for implementation of the process
- Resources required to implement the plan
- Quality goals and criteria for measurement

Quality Mission: The quality mission should be a part of the overall mission of the organization, rather than being separate and distinct. The organization's mission will embrace quality, apart from other things. As you know the mission is the purpose of the organization's existence. The mission will change only when the organization decides to pursue a completely new industry, a new market, or the same market in a completely different way. The quality, being an integral part of the business objectives, must be stated. It should be seen that the mission encompasses the new direction the organization has envisaged regarding quality.

Quality Strategy and Policy: A total quality strategy may embrace many objectives. These include:

- Improving service to the customers
- Improving business reliability and operating efficiency
- Improving or developing people involvement and commitment
- Improving organization-employee communications
- Establishing clear departmental goals
- Facilitating an open style management and team building
- Achieving the required accreditation

Strategic Perspectives

The important elements of a quality strategy are: leadership, people management, and processes. The behaviours of all managers must drive the organization towards total quality. A total quality approach should demonstrate:

- visible improvement in leading towards total quality
- recognition and appreciation of the efforts and successes of individuals and teams
- support of total quality by provision of appropriate resources and assistance
- involvement with customers and suppliers

The organizational processes consisting of value added activities are also important to the quality and Business Strategy. The processes that are critical to the success of the organization should be identified. Such processes should be systematically managed. The processes should be periodically reviewed for effecting necessary improvements. The individuals' creativity in process improvement should be stimulated.

The quality policy must be communicated throughout the organization. In a large organization it may be appropriate to develop divisional or sub-unit quality policies. The quality policy commits the organizational members to a certain way of doing things. The divisional policies must be complimentary to (and should in no way conflict with) the corporate policy.

Principles and Values: The principles and values create the environment in which everyone in the organization is enabled to work together to meet the organization mission and the quality strategy. These are basic to the creation of an organization culture. If the organization wants to change its culture, it may have to modify its existing principles and values. The management should examine the principles and values e.g., delighting the customer and seeking continuous improvement from the standpoint of its culture. The key question is how the organization will act towards its customers, its suppliers, and its employees. In today's business environment, the principles and values of a organization should also include its responsibility to the community at large, and in the case of a public organization its responsibility to all the stakeholders (including shareholders). Once the mission, the quality strategy and policy, the principles and values have been determined, they should be included as an integrated part of the quality awareness programme as educational material.

Organization and Processes: The management of organizational people is a critical element in quality management. How the skills and capabilities of people are developed and managed through recruitment, training and actual progression is critical to the success of quality strategy. The quality targets must be agreed to by the people and the performance continuously reviewed. The people should be involved in continuous improvement for which they should be empowered to take appropriate action.

What would be the organization strategy for TQM will depend on the size, structure and nature of the business? TQM organization is meant to be a facilitating process i.e. a process meant to assist for taking ownership of quality. It is meant to generate a process in which the concept of improvement becomes ingrained in each

individual. The TQM organizational process should provide for education, systems and tools so that all can take responsibility for quality. The TQM organization is not simply the responsibility for quality or quality improvement; it is more in the nature of a facilitating process.

TQM process starts with senior management and is cascaded down the organization. The Quality Steering Committee (QSC) is formed to focus senior management attention on the development of a strategy and plan for implementation. The organizational processes may start with the appointment of TQM coordinator who is responsible to the QSC for its success. The post requires the same level of authority as for all other senior operating executives. The title used for this position should fit the particular situation or culture of the organization. Vice President, or Director of Quality or Quality Manager are some typical titles. The central organization for TQM should be kept as small as possible. Any tendency to create a quality management empire should be resisted; otherwise it may alienate the line management and throttle the TQM process. The leaders must work through normal lines of the organization. The ideal approach perhaps may be for the coordinator to have a small secretariat. From time to time, a number of specialists may be added to the Quality Secretariat on permanent/ad hoc basis. The facilitators and others assisting the coordinator should generally retain their normal direct reporting lines within divisions, units or departments, with reporting via the TQM team organization to the coordinator. The teams thus are the normal essential management teams focusing separately on quality.

Educational plan: This is the heart of the TQM process. This part of the business plan must define the curricula and broad schedule of education and training required for each level of the organization. This is applicable not only for a organization establishing quality programme for the first time but also for organization which has quality programmes on an ongoing basis. The education-cum-training is meant to make the employees fully aware of the need for quality improvement and their role in the improvement process. The aims of the educational plan are to develop a common platform, to evolve a common language on quality, and inculcate shared values and principles. The perception of need and the ability to absorb new quality concepts may differ from level to level but the result must be a common commitment to continuous improvement. The training aspect of the plan is directed specifically at the individual's role in the organization or in the management of TQM process. Most training courses will contain both education and training elements.

The education plan may include a number of phases.

- **Phase I** may provide the executives and facilitators with sufficient knowledge about the plan embodying the improvement of process.
- **Phase II** may involve preparations for implementation. Members of TQM teams and task forces may attend process management courses. The senior and middle managers are usually concerned with this phase.
- **Phase III** may deal with the preparations for general education. A number of in-house instructors may be selected to teach the TQM course which is a central core of the organization wide education process. All managers/supervisors and specialized workers (e-g. engineers, software experts, etc.

may be required to attend a series of weekly two hours sessions). Communication will be strengthened if the instructors are drawn from management. This is usually not a full time role: instructors could teach one or two sessions per week. The objective of this course is to equip people to manage quality in their day to day operations.

- **Phase IV** may be concerned with educating the people i.e. workers at the floor level or clerical staff. This may be handled by the supervisors who had earlier attended the course in the previous phase (i.e., short supervisory workshops to prepare them for this role).

Tactical plans for implementation: The Quality Secretariat and the unit TQM teams may develop tactical plans for implementation of the quality strategy/policy. The tactical plans will be useful for monitoring the progress of the quality business plan and for maintaining the focus on team action.

Resources: The management should ensure that the resources required at each stage of the improvement process are identified, provided and properly utilised. The resources, may include financial resources and time; financial resources through external assistance, education and training etc., and time for individuals at each level of the organization. It should be made clear as to what resources will be supplied centrally and what from departmental operating budgets.

Quality Goals and Criteria: We have earlier emphasised that the quality plan should be viewed as integral part of the overall corporate business plan which establishes general business objectives and performance goals. Quality goals and criteria should be closely linked to the overall business goals. Two types of performance goals may be established in the initial phases of quality improvement. These can be defined as those designed to improve the quality performance of the organization and those designed to measure how well implementation of the improvement process is proceeding. Quality performance goals, as far as possible, should be quantitative. They could relate to the reduction of the cost of quality at stated periods, the defect levels rate and other ratios. Qualitative and quantitative criteria for measuring the success of quality will be derived from the quality policy and goals. The criteria would include goals for the completion of educational phases and periodical surveys and employees' attitude.

5.10 SUCCESS/FAILURE OF QUALITY STRATEGY/ PROGRAMMES

The Quality Strategy of any organization may fail. The reasons behind this may just be the same as those that cause corporate strategies to fail. Often corporate and quality strategies may come in each other's way. The organization may simply not be in a position to internalise the new strategy and therefore may fail to implement on a day to day basis. The concept of quality has to be organizationally embedded in a flexible, fast learning, high performance organization.

In the context of emerging competition the corporate strategic considerations,

for example, may be reduced to two key questions: how do we want to compete and where do we want to compete? The relevant/related questions could be: should our competitive difference be based on cost leadership or superior product concepts? Should we remain component suppliers or perhaps become a sub-supplier to a system supplier, or should we include more value added stages or development work in our business and market ourselves as a system supplier. The right quality strategy is absolutely an essential prerequisite for a successful business strategy. Deciding merely to take action in one area may be of little use unless the organization is in a position to tackle and take action on all matters.

Many organizations after having initiated or worked quality improvement programme abandon it later for one or the other reason. They are often not satisfied with the results achieved. The failures can be traced to one or more of the following main reasons:.

Lack of top Management's direction: Quality improvement programmes may fail because of lack of direction from the top management. Credible strategic objectives can be set only by top management. Even if objectives are set, the targets may not be concretised for individual assembly workers, sales staff or development engineers. Inputs rather than output ratios may be measured. Instead of registering improvements at the customer and over the entire value chain, records are prepared to show only how many teams are involved, how many meetings are held and how many ideas are collected. While this approach may ensure general enthusiasm at the outset, it may not yield any tangible results. The top management confines to initiating the quality plan without actively contributing to the implementation of the plan.

Lack of education and training: The quality programme in all probability will fail if employees have not mastered the quality improvement tools. The improvement in quality under such circumstances may either be unrealistic or difficult to sustain.

Over-bureaucratic setup: An over bureaucratic setup creates organizational barriers. Even highly motivated staff may lose its enthusiasm in the jungle of a bureaucratic organization. Change processes often fail because of an excess of hierarchy and functional interfaces which make decision paths too long. Even in a decentralized organization, barriers come to be created because the so called highly decentralised quality management function may administer the quality management programme in almost a commando style.

The successful TQM organizations overcome the above problems. They move the traditional functional organizations forward in following three dimensions:

- They set stimulating quality objectives top down;
- they mobilize the will and skills of employees towards continuous improvements bottom-up.

With simple targeted structure and processes, they create organizational conditions for flexibility and independent initiative.

Activity 3

Describe the elements of the quality business plan of the organization of your choice.

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How can a Quality Strategy Succeed?

In order that a quality strategy succeeds, a number of things need to be done. These are as follows:

Top management support: A quality strategy will be realised only after the corporate management gives its explicit support. A quality programme cannot be left solely to the mercy of the quality coordinator/ assurance manager. The top management's involvement in the formulation of quality strategy and its subsequent implementation is crucial to its success. The top managements in quality organizations are much more involved in developing the quality strategy. They play a more active role in its implementation. In quality organizations the top managements gives more than moral support. They take part in working sessions to keep abreast of individual topics and contribute to concrete solutions. After quality strategy has been formulated it is important to communicate it to all the segments of the organization and the people who have to work with it.

High level of aspirations, clear tasks and time scales: In quality organizations the top managements endeavour to create a climate of high achievement. Such organizations are often guided by global best practices and set targets based on them. They make an effective use of benchmarking. Clear tasks and time scales are laid down. Medium term objectives are laid down for the entire organization, which are further broken down into concrete sub-objectives for each function/ department/subunit. How these objectives are met should be systematically tracked. Right from the top manager through the shop floor workers, each individual knows exactly what target (dearly measurable) he has to achieve and by when, how this target fits into the overriding objectives and where he stands at present.

Tracking progress: If targets are long term, then milestones must be set over the period in question. This particularly applies to R&D which often has a very long term horizon.

Mobilising Employees: Quality organizations have carefully planned programmes that strengthen the 'will' and 'skill' of employees. They arouse enthusiasm of employees and improve their skills. Staff training for quality is of utmost importance. In addition to intensive training in quality tools, informal approaches which further develop the quality and cost awareness of employees play an important role. It may not be necessary to impart training on all conceivable tools

of quality. Training may be limited to those tools which are really relevant to a group of employees. If the workers know the techniques and accept them, they will have a sense of achievement when they see that consistent application of the techniques is producing better results. The quality organizations intensively train some selected employees who later function as missionaries or know-how centres.

It is also necessary to increase the cost consciousness of employees. Some organizations confront their staff with customers' reactions to quality. Sometimes, complaints may be put on the notice boards of workstations. Good organizations also put up on the notice board comments of praise from the customers. It is important to recognise the contribution of employees. Recognition acts as the strongest performance motivator. Personal recognition, approval and fair play are basic to employee satisfaction. Pride in achievement and perceived acknowledgement are the fundamental incentives for making improvements. Some hotels and hospitals have the practice of showing on the notice boards most enthusiastic or 'friendly employees' of the month. They invite suggestions from the customers or users of services for helping them in the selection of such employees.

Appropriate Organization Structure: A hierarchical structure is often not found suitable for an effective implementation of quality programme. The feedback system in a hierarchical organization is typically too long and too sequential. The decision making power is too fragmented. A flat organization is more meaningful for a quality programme. Team organization should take place at all the levels. The organization structure may have to be changed from the traditional functional structure to cost and profit centre based structure.

In quality conscious organizations there is more emphasis on the horizontal structuring rather than vertical structuring. There is emphasis on the core processes in the organization being generated by one source. The single source may be a group of people which produces a whole bundle of services. It takes full responsibility for the core process, for example, the entire manufacturing process from the receipt of raw material to delivery of finished products to customers. However, what a core process consists of may differ from organization to organization. For example, in an organization the R&D and production may be the two core processes.

Activity 4

Examine/assess the reasons for success/failure of quality strategy/programme in the organization of your choice. What could be done to make the strategy/programme a success if the experience has not been encouraging?

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5.11 SUMMARY

A Strategic quality management implies that corporate management recognises the importance of quality and regards it as a key strategic issue. Strategic quality management means aligning the organization and relating it to its environment and customers. A Customer satisfaction is the customer's positive or negative feeling about the value received from using an organizations product or services. The importance given to the customer does not conflict with the need of the shareholders or owners. A growing body of evidence shows that TQM, with its focus on customers, is the way to produce superior long term financial value for owners in a globally competitive market-place.

The business strategy of 'total quality' shows managers how to operate a business by focusing primarily on customers. The first and foremost focus of total quality is on consistently improving value to customers and thereby delighting them. The primary focus is the customer, and not the competitor as in competitive strategy. A distinction must be made between customer value strategy implied in TQM and competitive strategy. An organization guided by customer value strategy attempts to retain current customers as well as attract new customers by delivering best value to them. By doing so the organization tries to increase its market share.

In addition some other factors, TQM seems to be supporting the corporate strategy of concentration or related diversification. Many of the corporate restructuring exercises have been influenced by the strategy of related diversification.

TQM implies a radical change in the thinking about customers and suppliers. Customers are not mere buyers. A long term relationship between the organization and the suppliers is conducive for business growth. Shareholders/owners stand to gain from TQM programmes in the long run as such programmes enhance the value of the organization and have a salutary effect on the market share and stock prices. Since customer value is central to TQM approach, an organization must establish customer value determination. Customer feedback on what value the organization is giving to customers should be collected in a systematic way.

The organization for quality should fit with the structure and the nature of business. A quality business plan may include quality mission, strategies and policies, principles and values, organizational processes and communication, educational plan, tactical plans for implementation, resources to implement plans, and goals and criteria for measurement.

The main reasons for failure of quality strategy/programmes have been found to be lack of top management's direction, lack of education and training, and over bureaucratic set-up. It is therefore imperative that the top management supports the TQM programme evolves clear tasks and time scales, tracks the progress continuously and mobilises the employees, and evolves an appropriate organization structure.

5.12 KEYWORDS

Customer Delight : It can be created by exceeding the customer expectations.

Customer Satisfaction : The positive feeling about value received from using a organization's goods or services.

The Customer Value Strategy : A strategy that concentrates on exceeding the customer satisfaction, a strategy that recognises the centrality of customers.

5.13 SELF-ASSESSMENT QUESTIONS

1. Discuss the forces that have brought TQM in sharp focus over the past decade or so.
2. Can high quality and low costs go together? Illustrate your answer with examples.
3. What could happen if an organization concentrates on competitors at the expense of customers? How can it bring about a balance between the two sets of groups?
4. Can an organization pursue both the generic strategies of Cost Leadership and Differentiation simultaneously? Explain with examples.
5. What is meant by customer value? Explain the concepts of Value Realised, Value Sacrificed, and NetValue.
6. How can you determine customers' satisfaction? Explain.
7. What kind of strategy or strategic focus the TQM approach seems to be favouring or supporting and why? Justify.
9. What elements a quality business plan may include? Explain briefly.
10. Explain the reasons for failure of quality strategy/programmes. Cite examples/ case studies in support of your answer.
11. Discuss the imperatives for success of a quality strategy.

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UNIT 6 QUALITY CENTRED STRATEGIC PLANNING

Objectives

After reading this unit you should be able to :

- explain the interrelationship between quality and strategic planning;
- comprehend the concept of strategic quality management;
- understand the documentation systems and procedures regarding quality improvement;
- appreciate why it is imperative to achieve customer orientation.

Structure

- 6.1 Introduction
- 6.2 Role of Top Management in Quality Improvement
- 6.3 Strategic Quality Management
- 6.4 Customer Orientation
- 6.5 Quality Centred Strategic Planning
- 6.6 Strategic Planning Process
- 6.7 Summary
- 6.8 Key Words
- 6.9 Self-Assessment Questions
- 6.10 Further Readings

6.1 INTRODUCTION

We very well know that there is a link between Quality Management and Business Strategy. In the wake of increased globalization efforts quality has emerged as a strategic competitive tool. Even established business organizations these days are becoming prone to extinction due to stiff competition. In today's fiercely competitive world only organization using the best practices can survive and sustain. Earlier, quality was confined only to the manufacturing processes of the organization. With the advent of TQM quality has become all pervasive in many organizations.

It has been felt that top management of an organization should proactively participate in managing and improving quality. Hence quality needs to be integrated well with the strategy formulation and strategic planning processes of the organization. Quality is defined by the customer. Successful organizations attune themselves to the voice of the customer and customer satisfaction is not sufficient. Business organizations must strive to achieve customer delight and ecstasy. Towards the end of the unit we attempt to blend quality with strategic planning to

highlight the important role quality plays in strategic decision-making. In this unit we address several issues, in relation to quality and strategic planning, such as. Comprehension of these aspects will help you to adapt to different working situations.

6.2 ROLE OF TOP MANAGEMENT IN QUALITY IMPROVEMENT

Dr. W.E. Deming, the father of the quality movement strongly advocates that the quality concerns should be applied to management practice as a whole rather than simply considering them a domain of action of the quality engineer. He developed the concepts of Management Responsibility and intrinsic motivation of workers.

According to Dr. Deming 85 % of all defects are caused by the faulty management system and workers are responsible only for the remaining 15% of defects. Dr. Deming's most widely referred and quoted "14 principles of quality management" call for 'Elimination of numerical quota'. He further advocates that management should shoulder the responsibility of quality improvement and should develop an organizational climate in which workers can derive intrinsic motivation and satisfaction from the job well done.

Top management is a commitment to corporate resources. The chief executive is expected to provide his/her own time to the continuous improvement process. Most of the time the top management spends in defining the organization's vision statement, detailing corporate goals, outlining policy and planning for quality efforts and initiatives. The top management should create and lead the corporate level quality council in order to remove the barriers in the way of implementing quality in the organization. S/he should vocally support the total quality management movement in an organization. The top management must lead by example in demonstrating to others that quality is the top priority. The top management participation is defined in the following way:

"Which specific decisions are to be made by top managers?"

"Which specific actions are to be taken by the top management?"

Top management's personal participation in achieving quality improvement in all departments of the organization requires the following decisions and actions to be taken by it.

1. **Establish a quality concept:** The council works for directing and coordinating the organization's efforts to manage for quality. It establishes quality goals, providing the means against such goals and soon. In fact, strategic quality management is an extension of business planning by the top management. In large organizations quality sub-councils are established at the functional or divisional level. The task of top management is to coordinate the work of these sub-councils.
2. **Serve on the quality council:** The functioning of the quality council is severely affected if the members lack decision-making. Hence the top management should endeavor to serve on the quality council as that will

send a right message to the other members of the organization regarding the priority accorded to quality by the top management.

3. **Establish quality policies:** Policies are guidelines to managerial actions and they need the approval of the top management. The top management has to participate in establishing quality policies in the following ways:
 - Identifying the need for quality policies.
 - Assigning responsibility for preparing a draft of quality policy.
 - Reading, revising and approving the quality policy.
4. **Establish quality goals:** The quality goals at the top management level are very broad. Quality goal should be specific, quantifiable and time bound. At the top management level, the goal should be result-oriented rather than tool technique-oriented.
5. **Deploy quality goals:** Deployment means allocating the responsibility of accomplishing goals to lower levels of the organization. The lower level managers outline the specific actions to be taken to meet these quality goals and also the requirement of resources.
6. **Provide Resources:** It is very essential to provide resources for carrying out quality goals. Failure to provide adequate resources sends a wrong signal down the hierarchy throughout the organization that the quality is not a high priority area.
7. **Provide Team Oriented Training:** Training is a special type of resource which the top management is generally willing to provide. But the top management should be actively involved in identifying the needs for training in the field of quality in order to make it more effective and focused.
8. **Review Progress:** The top management should take upon themselves the task of periodically reviewing progress against the broad quality goals. This practice sends the right message regarding the priority given to it by the top management. The review can be done in the following two ways:
 - By receiving the summarized reports on actual quality performance (executive report on quality);
 - By auditing the processes of attaining quality (executive quality audits).

As a result of the review if the actual performance falls short of the goals the quality manager should discover and eliminate the root causes.

9. **Institutionalize Recognition and Reward System:** The top management should initiate action in institutionalizing a suitable recognition and rewards system. The top management should recognize and give rewards to meritorious individuals and groups for their excellent performance in the field of quality. They should organize and participate in ceremonial functions in which such awards are distributed and the specific employees'

contribution in quality improvement is publically recognized. Similarly the top management may decide to offer rewards in monetary terms or non-monetary terms.

Activity 1

Choose an organization and suggest what role the top management of that organization should play in bringing about quality improvement at all levels in the organization with particular reference to strategic planning and decision-making

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6.3 STRATEGIC QUALITY MANAGEMENT

While attempting to map the evolution of the quality movement in the United States of America, Garvin has described the following four distinct quality eras:

- Inspection
- Statistical quality control
- Quality assurance
- Strategic quality management

First three stages were necessary mainly because of operational considerations. The fourth one implies a new emerging vision.

Quality is no longer conformance to requirements but it is which enlivens the sense, instills wellbeing, and fulfils even the unexpected and unanticipated customer wishes. The new paradigm is Zero customer loyalty to other organizations/suppliers. Quality is to be treated as a strategic issue requiring priority and utmost attention to details. Quality is coupled with reliability to surpass customer satisfaction.

A systematic and methodological approach towards quality improvement shall help in instilling a culture that promotes excellence in all endeavors. A fundamental step in establishment of strategic quality management is that it should be very integrated with the strategic planning of an organization. The Box 6.1 shows the seven principles of Strategic Quality Management (SQM).

Box 6.1 Seven Principles of Quality Management

- **Proactive culture (Philosophy):** Early detection and counter measures of all the abnormalities.
- **Top down and bottom up approach:** It develops an employer-employee relationship where all are equal and part of a team focusing attention on development.

- **Scale every responsibility for quality:** Making people responsible for quality is less effective than creating a climate in which people are willing and able to take responsibility themselves.
- **Measure Cost of quality:** Cost of quality is measured by three costs: (a) Prevention (b) Appraisal (c) Failure. Prevention cost is the cost of getting things right the first time. Appraisal cost is the cost of checking and inspecting whether the things are done right. Failure cost is the cost of not doing it right.
- **Standards right first time:** It is costly to compromise on quality. Right first time is the standard by which quality standard is evaluated.
- **Scope:** Scope of strategic quality management is organizations wide. Quality improvement should be brought about in every functional area and department in the organization. It should not be confined to only the manufacturing department.
- **Theme of Continuous Improvement:** The business organizations should continually strive to achieve excellent levels in quality.

As is mentioned earlier quality objectives set out certain fundamentals and exhibit wide variations in nature and scope. Quality objectives may be required either for control or for the purpose of achieving some breakthrough in quality. Quality can be set organization wide or for departments. The scope of quality objective goes on widening, quality plans which are required to be developed for higher and higher levels. For example, when the objective is to establish and meet broad corporate quality goals, the organization needs to prepare a quality plan known as strategic quality plan.

Quality Planning Road Map

All types of quality plans are inter-related. Quality planning attempts to meet the quality needs of the customers. In order to meet these customer needs a quality planning road map can be prepared which is depicted in figure 6.1. The road map consist of structured and sequential steps. Output of each preceding activity or step becomes the input for the next step and so on.

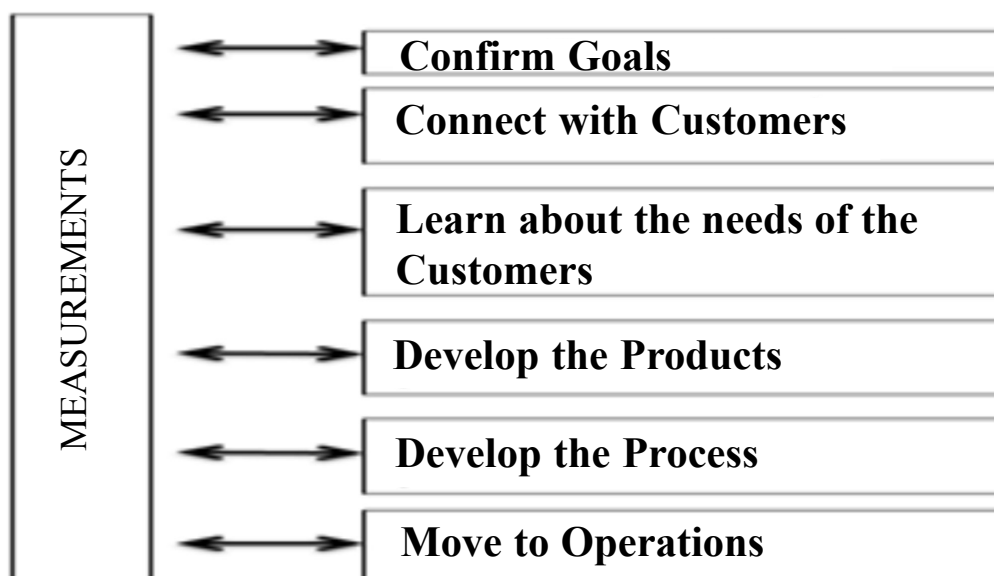


Figure 6.1: Quality Planning Road Map

This road map is universal in nature as it is applicable to quality planning throughout the economy, to all industries in both the manufacturing and service sectors. It is applicable at all levels in an organization, such as corporate, division department, and job and in all functional areas such as marketing finance, production operations and human resources.

The quality planning road map can be applied at the following levels:

- a) Supervisory and worker level
- b) Functional level
- c) Multifunctional systems
- d) Major programmes

a) Supervisory and Worker Levels:

Each employee is assigned a job. Large number of jobs requires quality planning as well as re-planning as the job may be dynamic. In some cases the employees themselves are given training to plan their jobs. Some of the concepts and tools which are used for quality planning at this stage are:

- (i) Self-control concept: A person is said to be in the state of self-control, when work job is organized in such a manner so that the person can have full control as well as mastery over the attainment of planned results. Such a person can be held responsible for the performance results. In order that a person can achieve self-control s/he should be provided with:
 - knowledge of what s/he is supposed to do;
 - knowledge of what others are doing; and
 - Means for regulating either of the two above if there is failure to meet the objectives.
- (ii) The triple role concept: Each employee holding a job should be able to carry out the following three roles:
 - As a customer
 - As a processor
 - As a supplier

This concept is depicted in figure 6.2. Using this concept natural progression of quality planning can be achieved.

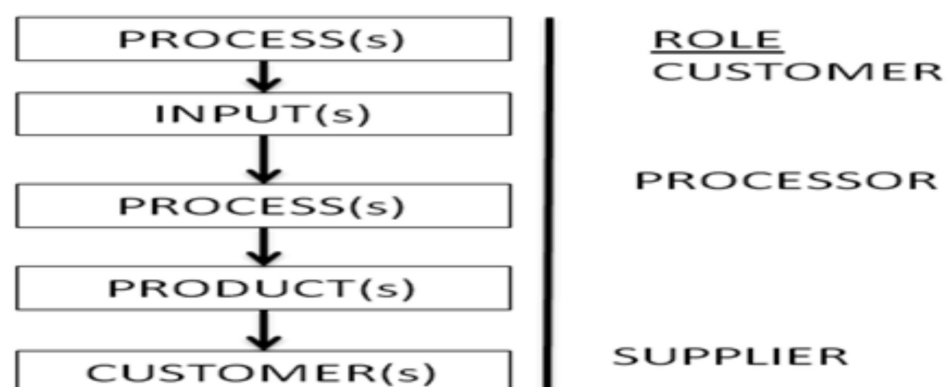


Figure 6.2: The Triple Role Concept

b) Functional Level

Functional managers in charge of different areas undertake quality planning to manage their performance in a better way. For example, planning for the marketing function places major emphasis on the defined role of quality planning in marketing department. In most cases the functional managers are not trained professionals in the field of quality. Hence, either they can receive help and guidance from quality specialist or they themselves can be trained in quality issues to become quality professionals. Training may be given in the field of skills and tools of planning for quality.

c) Multifunctional Systems

Some systems such as Management Information Systems (MIS), Human Resources and New Products Development are all pervasive. These systems have spread their tentacles in many functional areas as they receive inputs from all major functional areas. Inadequate quality planning may jeopardize the working of the systems. Central quality planning is highly useful in achieving coordination among all the functional areas involved. Various methods are available for achieving central planning with emphasis on quality:

- A team (or committee) of functional managers.
- A team (or committee) of functional managers, with the assistance of quality specialists.
- A project management department.
- The quality assurance department.

d) Major Programmes

The term major programme is used here in the context of costly projects such as nuclear power plants, development of a new weapon system, launching a satellite into space etc. Each of these projects is so complex and unique that they require specific quality planning for the major programmes in question. In these programmes it is common for clients to specify the quality plans and also the product project performance specifications. For meeting these specifications the organizational structure used by the contractors (i.e. suppliers of these programme projects) should be the matrix type. Moreover, the contractor should provide assurance to the client that all the specifications have been met as per the contract. The documentation part of the contract may be very intensive. In addition to achieving the quality assurance, it may be helpful in the event of legal battles.

Strategic Quality Management (SQM)

The major application of quality planning concept is in Strategic Quality Management (SQM). (SQM) is an extension of organization's business planning to include quality planning. The sequence of activities to be performed in strategic business planning is as follows:

1. Establish vision
2. Set up mission statement

3. **Determine the objectives/goals**
4. **Identify the strategic alternatives**
5. **Choice selection of strategic mix**
6. **Implementation of strategy by allocating resources.**
7. **Evaluation of the strategy**

In SQM the same sequence applies to managing for quality which takes the following form:

- Establish broad business goals;
- Determine the actions needed to meet the goals;
- Organize and assign a clear responsibility for meeting goals;
- Provide the resources required to meet those responsibilities;
- Provide the required training;
- Establish the means to evaluate actual performance against goals;
- Establish a process for periodic preview of performance against goals
- Establish a reward system which relates rewards to performance;

The SQM requires special organization design, skills, tools and training. The most important facet SQM is the involvement in top management quality control.

Purpose of planning for quality is very important to be achieved. Planning for quality has been delegated to the functional departments who in turn are responsible for establishing and meeting quality goals specifically set for the respective functional areas. Due to lack of coordination among various functional departments this approach has not succeed in many organizations. The application of SQM has not only contributed in the betterment of quality but has also resulted in improved performance in many organizations.

The advocates of SQM point out that by implementing it, the concept of quality can be extended to all business processes of the organization beyond the manufacturing department, thus leading to total quality management. A more cohesive and well-knit team results due to the application of SQM because the harmony between top management and the functional manager increases. However, the adoption of SQM leads to increase in the workload of top managers who are already very busy. The functional managers may perceive the SQM initiatives as the invasion on their autonomy. This problem is more than compensated for by the fact that, due to SQM, valid and viable quality plans of the functional managers receive adequate resources from the top management as they themselves are involved in the quality initiatives and possess a better understanding of quality-related issues.

Deployment of Strategic Quality Goals

It was earlier stressed that the concept of strategic business planning should be enlarged to Strategic Quality Planning (SQP). In SQP broad quality objectives

are established to cover the whole spectrum of an organization. These quality objectives should then be deployed to lower levels of management who in turn set up functional level quality plans to identify the actions needed to achieve the quality objective and also the resources required. The methodology of SQP starts with a focus on the 'driving forces' to establish strategic quality objectives (or goals). These driving forces have a significant impact on quality. These forces are the external environmental factors such as technological, social, political and economic factors. Other factors of the relevant environment include competition, suppliers, specific technology issues (state-of-the-art technology) and internal corporate factors such as core competencies in the areas of critical success factors. The top managers should evaluate these forces and their possible impact on quality by collecting suitable data and analyzing it to yield meaningful information.

Broad and strategic quality objectives should be deployed to lower levels of management. Such deployment requires both upward and downward communication. Strategic quality goals may be established and proposed by the top management. The lower level (generally divisional functional level) managers identify the action plans which will achieve these goals and they also submit the resource requirement plans to the top management. Both the top managers and functional managers negotiate to arrive at an optimum balance for achieving the quality goals as well as the cost of doing so. Strategic goals are often established as a result of nomination from the lower level managers. They may identify quality-related needs for which the strategic quality planning is required as the top managers support is essential due to:

(i) extensive resource requirement; (ii) its impact on multiple divisions functions; and for setting major precedents which are required.

It is therefore essential that business organizations should have a synergistic combination of both strategic and operational planning.

Activity 2

State the quality policy and quality objectives of an organization of your choice. To what extent, in your opinion, is the functioning of your organization able to achieve quality policy and objectives? Discuss.

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Planning for Strategic Quality Control and Improvement

Planning precedes all control activities. A planned system of quality control is used to ensure that the organization has not lost track of the original objectives.

The quality control process consists of the following steps:

- Choosing the control subjects.

Strategic Perspectives

- Choosing a unit of measure.
- Establishing quality goals for operating performance.
- Creating a sensor (a means of evaluating actual performance in terms of units of measure).
- Evaluating actual performance.
- Interpreting the difference between actual performance and quality goals.
- Taking corrective actions on the difference.

The above steps take place within a system called feedback loop as shown in figure 6.4. The loop starts from the control subjects (i.e. process) which is to be regulated. A sensor measures the actual performance while goal is established as the desired level of the control subject. The collator's task is to compare goals with the performance and if significant difference persists the actuator is used for making adjustments in the process, or sometimes the goals themselves may have to be adjusted in order to achieve parity of performance with the goals/resulting in a closed feedback loop. Modern production/operations departments make use of computerized online systems of quality control.

Quality improvement is directed largely at creating beneficial change. Planning quality improvement can take the following two major forms:

- (i) Improvement in product features to provide customer satisfaction. It can be achieved by applying the quality planning road map. The quality improvement results from re-planning (if necessary) about quality-related issues.
- (ii) Improvement to achieve defect free production to minimise the wastages and increase in the productivity.

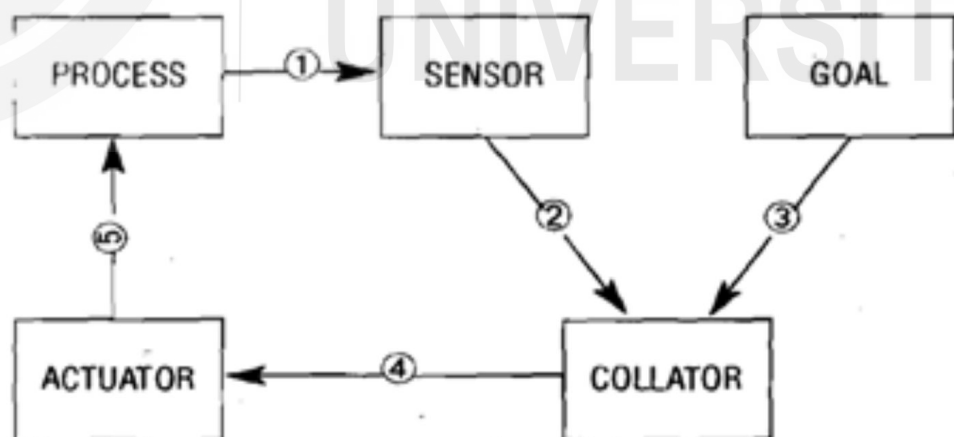


Figure 6.4: The Feedback Loop

6.4 CUSTOMER ORIENTATION

It is very important to become attuned to not only what the customer demands of today's products and services but also what the customer will want in future. One of the corner-stones of the quality approach is to identify and meet customer need. Customer, in its broadest sense, encompasses both internal and external,

consumer and industrial, products and services purchaser and user, intermediary and final. Consumer, in other words the customer includes everyone impacted by the product processes. Listening to the customer is the term most often used in quality discussion.

Achieving Customer Orientation

Theodore Levitt (1975) was the first one to argue that an industry was a customer satisfying process, not a goods producing process. Being customer-oriented is critical to the success of any business or industry. Customer orientation, first and foremost, is a business philosophy which focuses outwards and dements agenuine commitment to satisfy customer needs and wants. Without such outlook, focus and even obsession the customer orientation cannot be achieved.

The process of achieving customer orientation is briefly described below. It is necessary to instill customer orientation where it does not exist. It is recognised as a process of organizational change which has to be introduced in a planned manner. Figure 6.5 is a fishbone model that specifies the steps to achieve customer orientation.

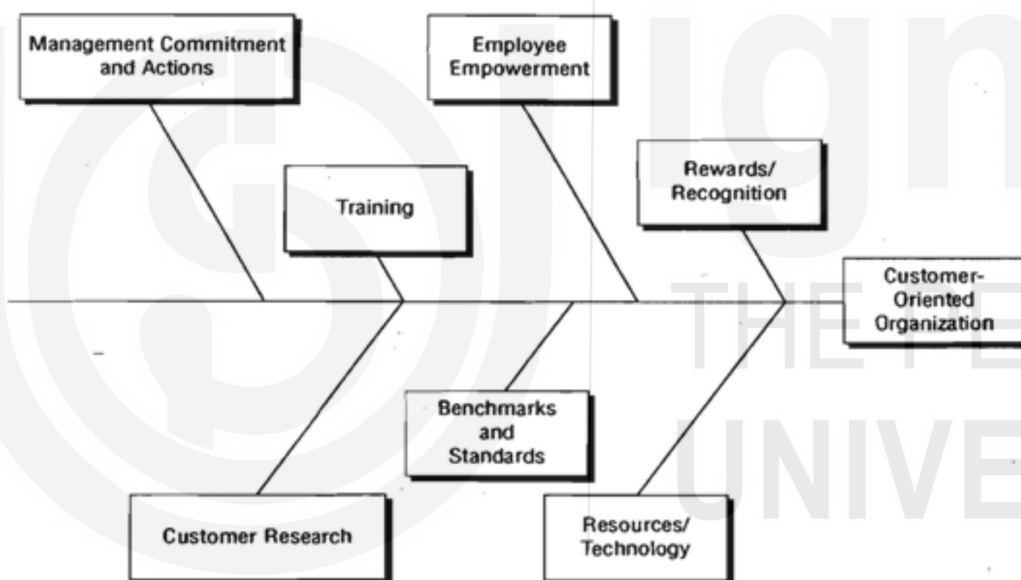


Figure 6.5: The Process of Becoming Customer-Oriented

The steps involved in the process of organizational change are as follows

- (i) Management commitment and actions
- (ii) Training and customer research
- (iii) Establishment of benchmarks and standards
- (iv) Employee empowerment.
- (v) Resources/technology.
- (vi) Rewards and recognitions.
- (i) **Management commitment and actions:** The top management should be genuinely committed and demonstrate it by their actions that they assign the highest priority to achieving customer orientation. They should be actively

involved in developing an appropriate customer complaint system which will obtain customer feedback in future. The top manager should ensure that information about customers is available and is being used in decision-making.

- (ii) **Training and customer research:** The employee should be trained in customer research techniques in addition to training in group dynamics and problem-solving techniques. Some sophisticated techniques such as quantitative analysis of customer databases require help from the quality specialists. Some selected middle-level managers may be imparted these skills through training. Customer research should be conducted to identify customer needs and wants. It is a perplexing problem to identify future customer wants as the future technology may not yet have been unfolded. In these situations Insight generating techniques and their creative applications are more helpful than mere quantitative analysis.
- (iii) **Establishment of benchmarks and standards:** Benchmarking performance against best practices is a powerful stimulus for improvement. It can be applied to the area of customer grievance handling mechanism of an organization. Standards are objectives that are established on the basis of customer expectations and behavior. Standards can be set only when the customer expectations have been measured. It is left to the wisdom of management to decide whether they should simply satisfy customer expectations or they should strive to exceed customer expectations to achieve customer delight.
- (iv) **Employee empowerment:** Empowerment implies pushing decision-making down to the lowest possible level. It means giving team's power to make decisions which may satisfy customers. It is difficult to practice as it implies relinquishment of decision-making power by management which all along has been their prerogative.
- (v) **Resources/technology:** Employee empowerment alone is not sufficient. They should have other organizational resources which may involve the use of the latest technology to satisfy the customers. However technology alone cannot solve a customer's problems. It requires a human intervention to show concern and judgment while serving customers.
- (vi) **Rewards and recognition:** Rewards and recognition should be used to acknowledge both increased productivity and also success (less measurable) in satisfying customers. Each and every employee involved in the quality process should be publically recognised in celebrations in which the top management should also take keen interest. The recognition can take any of these forms: a certificate, recommendation letter, a training offer, or something more creative. Quality planning should treat rewards and recognition as important strategic issues. These are the necessary activities to achieve customer orientation. The customer should always be at the centre of both planning and execution. It has amply been stressed that becoming and remaining customer-oriented requires an organization-wide commitment. The entire organization right from the top management down to the organizational hierarchy up to frontline personnel (who are

in direct contact with the customers) must recognise the importance of keeping customer satisfaction and delight as a central focus. More importantly their actions must consistently put customer satisfaction first. Then only can an organization truly become customer-oriented.

Activity 3

Assume that you are working in an FMCG Organization.

- a) Do you know who your external customers are? In what ways has your performance a bearing on satisfying and delighting your external customers.

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- b) If the concept of internal customers is practiced in your organization, what conscious efforts are taken by you to satisfy your internal customers in the organization?

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6.5 QUALITY CENTRED STRATEGIC PLANNING

An organization needs a vision and a strategic plan that embraces quality as its foundation. The increasing quality consciousness gives rise to the following philosophical and practical issues for the organizations:

- Is quality a means for achieving the organization's mission or is quality an integral part of the mission itself?
- Why should quality be integrated into the organization's overall strategic planning process?

It is not possible to separate quality from the organization's core mission and purpose. It is a wrong notion to treat it separately just as a strategy for achieving the mission. No organization can survive for long without placing more emphasis on quality. Malcom Baldrige National Quality Awards include in its list of criteria the concept of having a strategic plan to implement the quality process of an organization. Without strategic vision and plan, an organization would not be able to implement its efforts in quality improvement. J.M. Juran advocated the increased participation of senior management to proactively plan and manage for quality. He suggested that the top management should have a strong focus on quality as their core responsibility.

The organizations today need a process for strategic thinking and planning that embraces quality as the organization's core value. As a result of this understanding of foundational role of quality in strategy formulation, the term strategic quality

planning was changed to strategic planning in Malcom Baldrige National Quality Awards in the year 1996. It indicates that the quality should blend well with the organization's philosophy and actions.

All organizations establish objectives (or major goals) which must be achieved in order to accomplish the mission. Enabling actions are strategies and tactics that an organization employs to achieve its objectives. For example, an organization's mission may be to provide high quality health care services. It can have the objective to always provide the right medicines to the right patient. For achieving this objective, an enabling action will have to be a quality control programme which will support the mission of the organization. Specific quality tools such as audits control charts are some of the enabling actions which can be thought of as a means (strategies) to an end (objectives).

It is clear that a strategic plan is a planning document, developed through a specific process that orients an organization towards its mission by placing its central focus on the objectives. It enables actions that impact quality. Figure 6.6 shows the continuum of perception about quality.

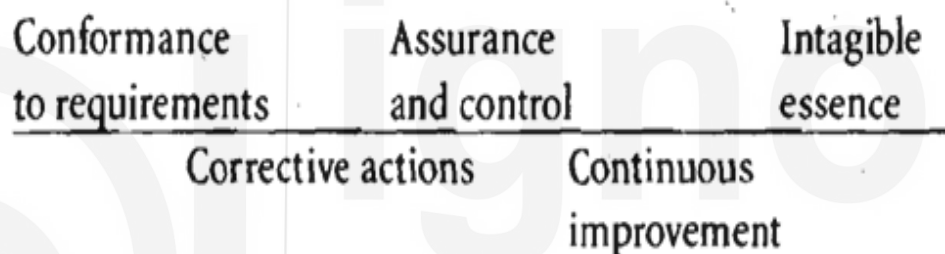


Figure 6.6: Continuum of perceptions about quality

How does an Organization Relate to Quality?

The point of view an organization has about quality will influence the manner in which it relates to the strategic planning process. There are five perceptions about quality which are shown in the figure 6.6 while the first three do not include quality as an important component of the strategic vision. The last two treat quality as an essential ingredient. The first three approaches of quality are based on reactive thinking styles about the future and the last two approaches namely; continuous improvement and intangible essence are proactive strategies. We shall briefly discuss the last two approaches to quality:

Quality as Continuous Improvement

Quality is a continuous improvement process. Using this approach improvement is brought about in every system of the organization by removing errors and achieving breakthroughs in performance. Juran is credited with the origin of this concept, in which teams and individuals focus on specific opportunities to achieve a breakthrough. This concept helps in aligning the quality functions with the strategic planning process. Continuous improvement in quality helps in shaping the destiny and influences the future of the organization.

Quality as Intangible Essence

In this approach quality is viewed as an intangible essence rather than being a quantifiable ingredient. Quality is being seen as a standard of excellence which is

difficult to define. The perception of quality as an intangible essence can impact the strategic planning process of an organization. This approach means that there is an unquantifiable spirit or commitment to excellence in an organization.

Quality and Strategic Planning

Figure 6.7 shows the dynamic link that exists between perceptions of the future (shown on horizontal axis) and the perceptions of quality (on vertical axis). They are shown on a scale which ranges from reactive to proactive. As a result of various combinations four possibilities emerge:

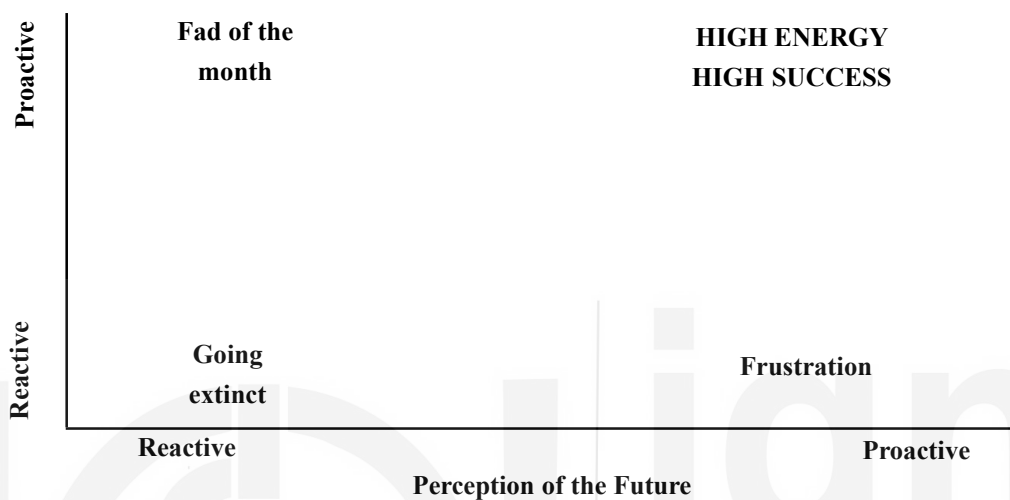


Figure 6.7: Perception of Quality and the Future

(a) Fad of the Month

If organization is proactive in quality but has reactive perceptions about the future they will introduce one fad after another to practice the latest quality tools but with little effectiveness.

(b) Frustration

A reactive approach to quality when combined with the proactive approach to the future leads to high frustration. The organization may have great plans for quality but they don't find an opportunity to implement the same as they are involved in fire-fighting or managing one crisis after another.

(c) Extinct

A combination of reactive quality as well as future may endanger the very survival of that organization which may become extinct.

(d) High Energy High Success

An organization can become dynamic and thrill its customers (also capture the market share) if it combines a proactive approach to quality with proactive thinking about the future. This combination leads to a kind of a fission reaction in which when two highly potent masses are placed together a lot of organizational power is unleashed to generate high energy which will lead the organization to achieve high levels of success.

Activity 4

In the light of the discussion in this section, what suggestions would you give to the top management of a service organization, if asked by them, in achieving quality orientation in the strategic planning and management of the organization?

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6.5 STRATEGIC PLANNING PROCESS

In order to integrate quality with the strategic planning process a systematic and sequential procedure has to be adopted.

The following nine broad steps comprise the ‘strategic planning process’.

1. Listen to the customer’s needs.
2. Plan how you will meet the customers’ needs.
3. Aligning the organization with the values.
4. Awareness about the influencing forces.
5. Develop specific quality objectives.
6. Consider various scenarios.
7. Plan to close the gaps.
8. Take actions to achieve the objectives.
9. Reevaluate and renew efforts.

1. Listen to the Customer's Needs

The basis of strategic planning is the identification of customers and their wants and needs. An organization must seek its customers’ requirements, expectations and assess future trends before developing a strategic plan. Customer requirements include price customers are likely to pay in addition to reliability and performance. Frequent listening to the voice of the customer shall provide the organization much useful customer feedback. Viewpoint is to treat customers as stakeholders in the organization. The quality policy should be relevant to the expectations and needs of its customers.

2. Plan to meet the Customer’s Needs

The organization must determine its positioning with regards to its customers. Various alternatives such as whether the organization should give up, maintain or expand market position in some or the other product categories should be

considered. An organization in order to become successful should concentrate and consolidate its position in the areas of excellence to be able to provide high quality products services to its customers in the most profitable manner. This can be achieved by formulating the mission statement of the organization which will guide it in delighting the customers and clients. The mission statement with emphasis on quality provides legitimacy to the quality in the organization. Employees are encouraged to believe that quality improvement is a core part of their jobs. The focus on quality in its mission statement will help an organization to come closer to meeting and exceeding the customer expectations.

3. Awareness about the influencing forces

The next step is to develop the internal vision of the organization after the organization's positioning with its customers has been clarified by developing a mission statement with strong emphasis on quality. Sometimes the organizational restructuring has to be attempted to achieve the desired outcome. Once the mission statement is prepared developed the organization need to invest some more time in reflecting on their vision and values before developing a strategic plan. The strategic plan provides the path for achieving the vision and mission of the organization. The value statement provides guidelines for work ethic and the behavior of employees. The values define what people in the organization believe. For a successful strategic plan the following core values are essential:

1. Drive out fear.
2. Participative leadership.
3. Use of data and management science techniques.
4. Continuous improvement.
5. Commitment to learning.

The vision statement (or a set of value statements) describes where the organization wants to be in terms of “overall performance and competitive leadership”. This is the core of the quality centred strategic planning.

4. Awareness about the influencing forces

Strategic planner must devise an appropriate mechanism through which they can scan both the present and future trends of external environment that will exert an influence on the organization. A review of an organization's history helps in discerning future trends. The success of strategic planning process depends on the ability to think of the future as an environment that is shaped and formed by plans and actions undertaken by organizations in a proactive manner. ‘The Force Field Analysis’, as shown in figure 6.8 a technique developed by Kurt Lewin, a social scientist is useful in the scanning process. This technique visually displays the various positive and negative forces that impact a situation.

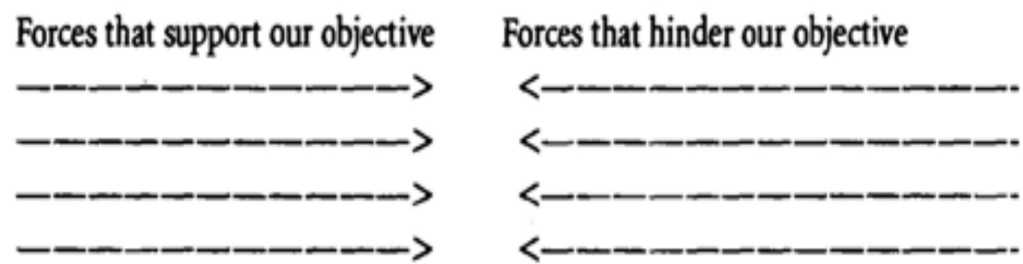


Figure 6.8: Force Field Analysis

5. Develop specific Quality Objectives

At this stage it is imperative to identify the main gaps between where the organization is and where it wants to be. Specific objectives will be identified by the strategic planning team on the basis of Force Field Analysis as well as by analyzing its internal core competencies. Answering the following question helps in identifying the specific objectives for achieving the specific plan of an organization:

“What are the objectives we need to accomplish to achieve our mission and vision based upon our values and the forces impacting our organization?”

Several broad objectives can be identified such as:

- (a) Lowering production cost;
- (b) Achieving operational excellence and
- (c) Constantly improving standards of quality.

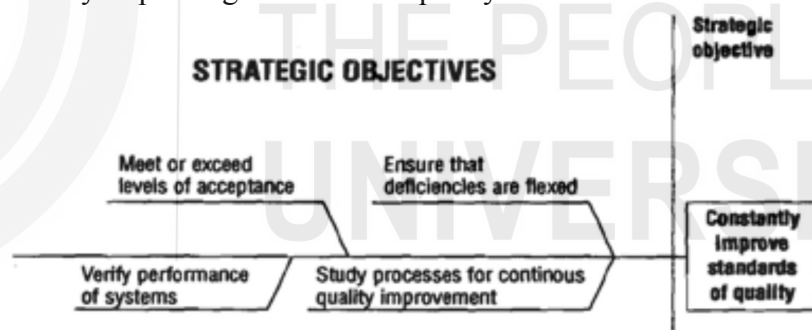


Figure 6.9: Actionable Key Business Drivers

All these three are vital to achieving the mission and vision. The key strategic objectives are also called the actionable key business drivers which are shown in figure 6.9.

6. Consider Various Scenarios

In this step various areas are considered which can be envisioned on the basis of three possibilities:

- (i) Events going well for the organization;
- (ii) Events not so well for the organization;
- (iii) Events going in a manner that is most likely to happen.

Using multiple scenarios offers many advantages. Based on the mindsets of the strategic planners and thinkers either the optimistic, pessimistic or middle-of-the-road scenarios can be developed. It enables an organization to prepare itself to be flexible in adapting to any of the above scenarios. It will also help in identifying gaps that exist between where the organization presently stands and where it needs to be.

7. Plan to Close the Gaps

The strategic planning team identifies and establishes goals which will assure high quality in their endeavors. They should perform gap analysis to find the difference between current situation and the goals. Quality improvement is not possible without a specific plan to close the gaps. Assessing the relative importance and relative difficulty of each gap enables strategic planners to direct their energy in closing the gaps. This may also be termed as process improvement. Organization should embrace quality as an essential ingredient in their vision, mission and objectives. They should also increasingly use specific statistical quality tools to achieve their quality objectives. Figure 6.10 highlights some tools.

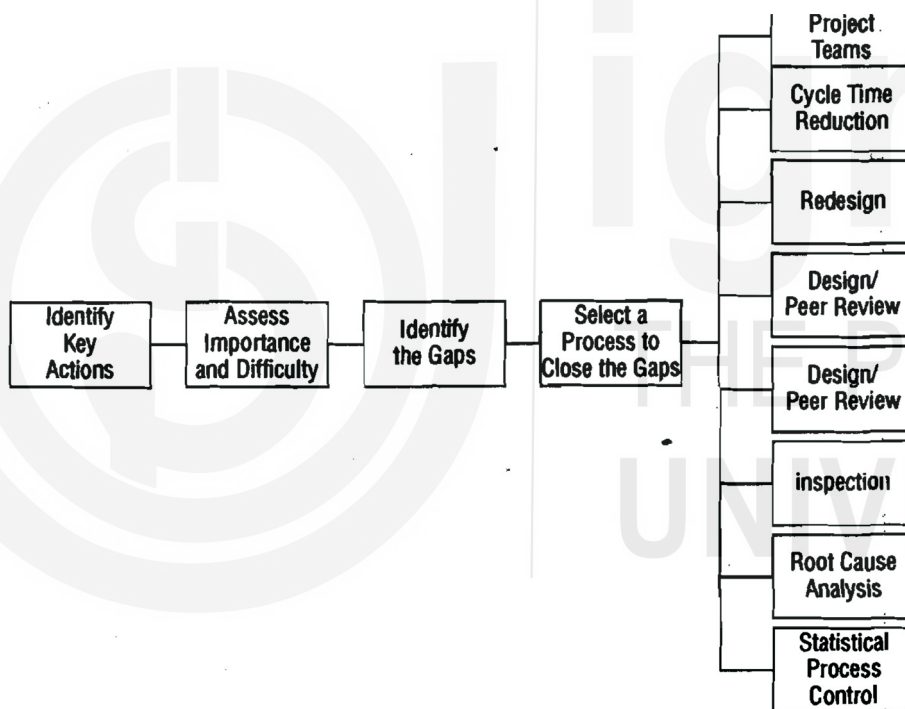


Figure 6.10 Quality tools to achieve Quality objectives

It is necessary to use the right tools to help the organization close the quality gaps. The strategic planners must take some steps to close the gap between the vision and current reality.

8. Take Action to Achieve the Objectives

At this stage the time is ripe to get down to the implementation of action plans and also to develop a suitable mechanism for tracking the performance results. Issues given below are of paramount importance:

- Should the organization focus on measuring the implementation of the action plan?
- Should it focus on the objective in the strategic plan?

The use of an external standard for benchmarking performance can be a useful technique. Such standards can help in deciding about the implementation of a strategic plan in any organization. One of the criteria that can be used is customer satisfaction. Customer satisfaction is the biggest proof that the customer's inputs are well integrated with various organizational processes and activities. If customers are not satisfied then it is clear that the organization is not on the right path in its pursuit of strategic quality objectives. This consciousness leads management to take corrective actions to improve the situation. The management should develop an action plan for every gap identified in the strategic plans. It defines what actions will be taken by whom and when. These action plans should be developed with a commitment for implementation by line managers.

8. Reevaluate and Renew Efforts

The external environment is continuously changing, causing an impact on the organization. A strategic plan once developed is not static as it has to come up to the expectations of the dynamic realities. The strategic planners should reassess and reenergize the system before the energy and enthusiasm wanes in the earlier established strategic plan. Otherwise the strategic plans will become obsolete and forgotten.

In order to keep the organization focused on the path of quality improvement, it is very important to think of renewing strategic plans in the light of prevailing environmental conditions. There may be some resistance to the change by line managers as they have to reframe their action plans.

The need and importance of the process of continuous quality improvement should be explained to the order to manage resistance to this change. The strategic planning process connects all of the components of the quality programme to achieve high levels of quality and to satisfy their customers which will enable them to survive and prosper in future.

6.7 SUMMARY

At the beginning of this unit a conceptual framework was developed for briefly introducing the concepts of strategy and strategic planning. The top management plays a crucial role in strategic decision-making and without its conviction it is practically impossible to achieve the quality strategic planning focus in any business organization. Top management should take a keen interest in developing the quality policy and objectives and should provide adequate guidelines to the quality managers to help them in preparing quality manuals. These documents are corner-stones of quality management system of an organization. As the success of an organization in achieving accreditation certification from any agency (like International Standards Organization (ISO) or Malcolm Baldrige National Quality Awards Committee) hinges on their accuracy and adequacy.

In order to accomplish the milestone of Total Quality Management an organization has to plan its journey according to the quality planning road map which is applicable at all levels in an organization. The process of setting up strategic quality goals and their deployment up to the lower level of an organization has

been discussed. The role of planning in exercising quality control and through it quality improvement has also been outlined.

'Delighting the customers' continuously is the aim of many business organizations. It requires carefully listening to the voice of the customer and acting accordingly. A fish-bone model was described which specifies the steps to achieve customer orientation. It can be used to carry the voice of customers to all parts of the organization with the help of cross-functional teams. The last section of the unit, devoted to the issue of quality strategic planning, emphasize the need for following proactive styles of thinking about future. In order to better understand the dynamic link that exists between the styles of thinking about the future and perceptions of quality, the two have been plotted together in a graphical manner to yield four possible strategic alternative courses of action. Finally, the nine steps strategic planning process with quality as its core focus has been described in detail.

6.8 KEY WORDS

Customer delight	: Exceeding the expectations of the customer by providing very high levels of quality and pleasantly surprising the customer.
Quality strategic planning	: A strategic planning process in which quality is embedded in each and every step.
Quality function deployment	: A structured process that carries the voice of the customer to all parts and processes of the business organization.
Quality manuals	: Elaborately thought out, formally approved and authorized written document in the published form.
Quality policy	: A well-documented policy statement clarifying the role and importance accorded to quality in an organization.
Strategic planning	: It is concerned with direction and focus of future activities in a planned manner.
Strategic quality management:	Embodies a dramatic shift in perspective, where quality is treated as the essential ingredient of the strategic management process.
Strategy	: The broad programme for defining and achieving an organization's objectives and implementing its mission.

6.9 SELF -ASSESSMENT QUESTIONS

1. Are quality and strategy inter-related? Discuss the role of quality in strategic business planning process of a business organization.

Strategic Perspectives

2. What are the roles and responsibility of top management of an organization in achieving quality centre strategic planning focus for an organization?
3. Briefly describe various sections of a quality manual. How can it be used to achieve quality improvement?
4. What do you mean by quality planning road map? Can it be applied to the major programmes at corporate level? Justify.
5. How can strategic quality goals be deployed at various levels in an organization?
6. “Quality is consistently achieving customer delight”. Comment on the statement.
7. What is Quality Function Deployment? How can it be used to translate the voice of customer for various business processes of an organization?
8. What does the “Quality centred strategic planning process” mean? Briefly describe the nine-step-procedure of strategic planning.

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